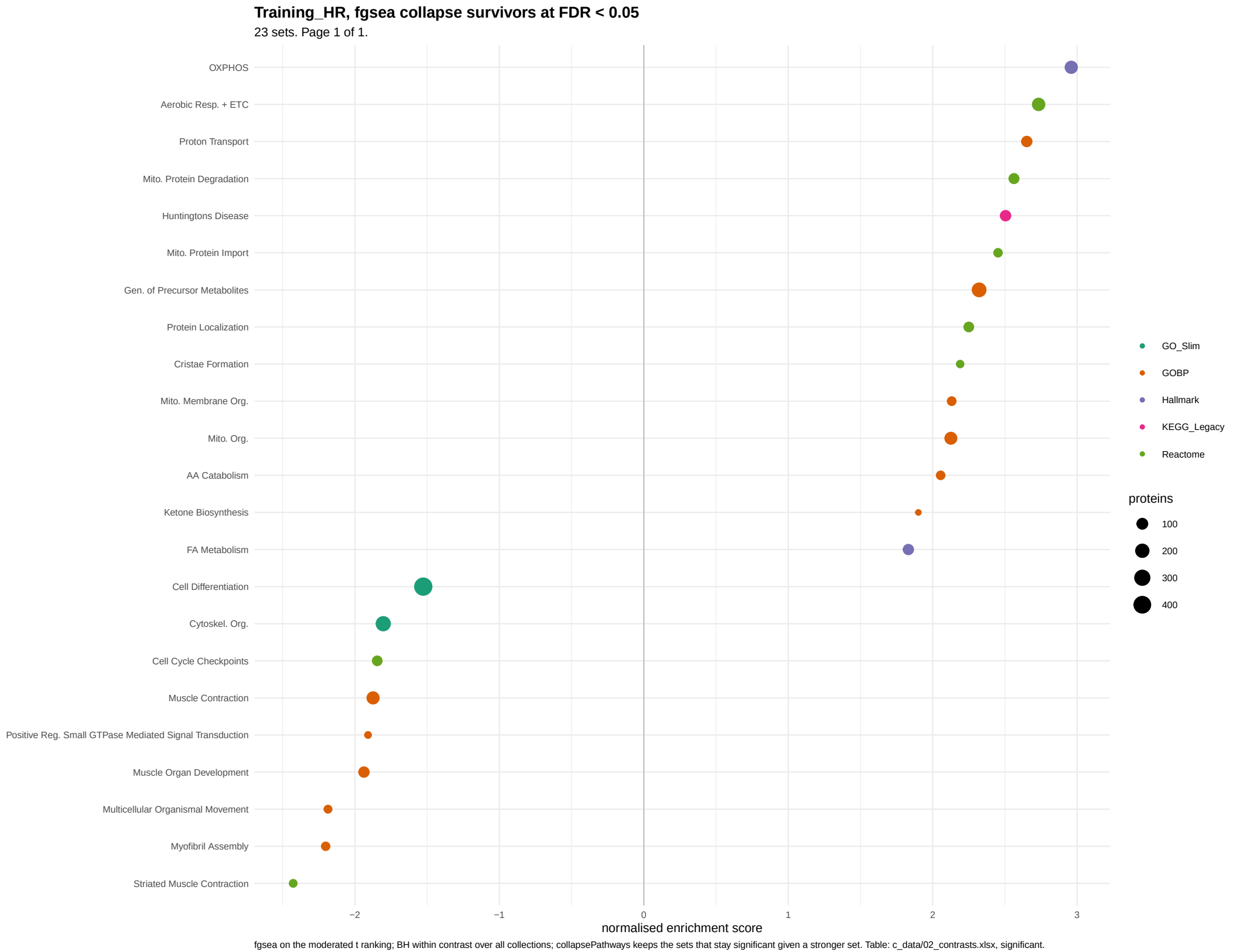
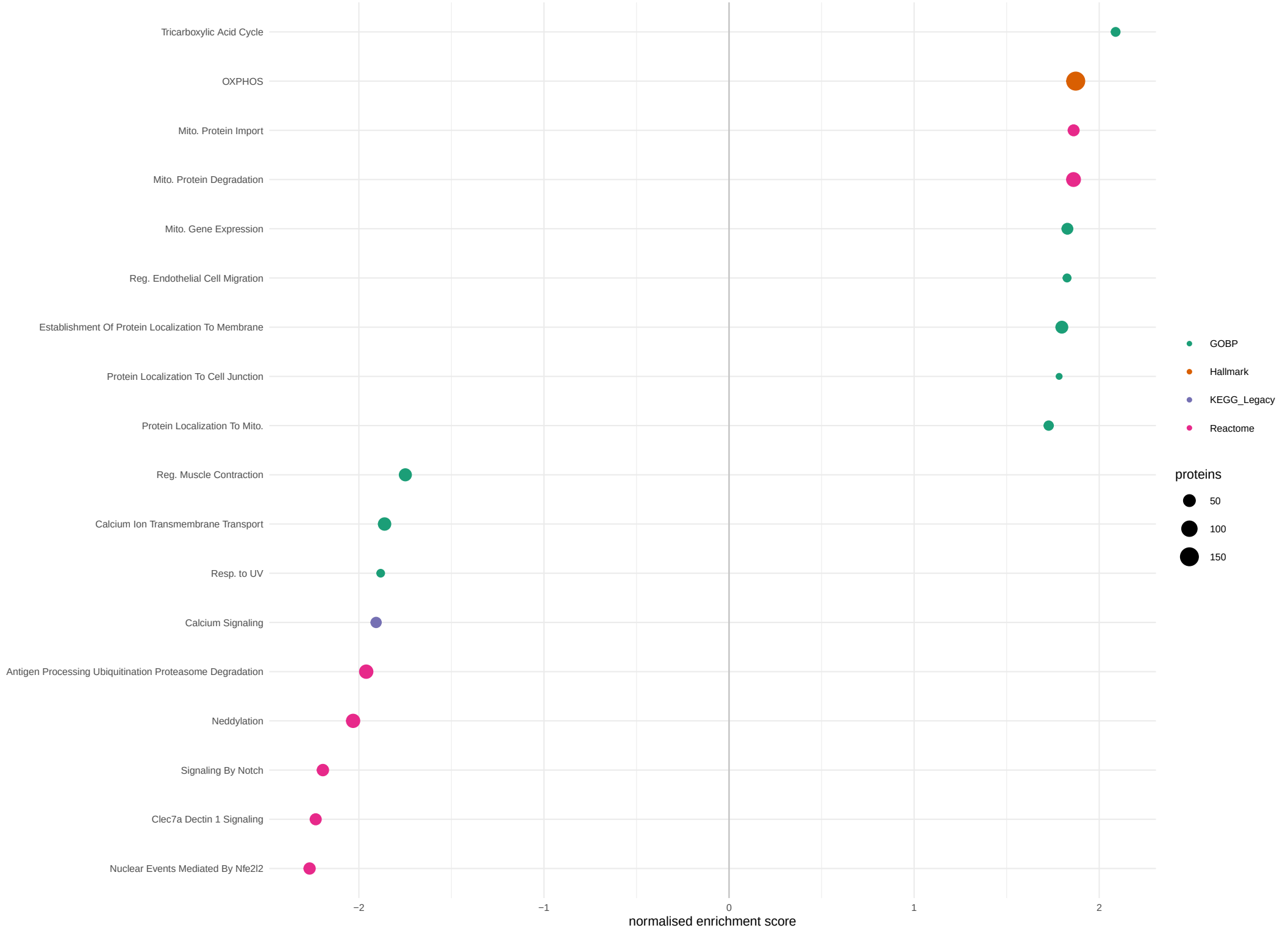

Training_HR, fgsea collapse survivors at FDR < 0.05
23 sets. Page 1 of 1.
OXPHOS
Aerobic Resp. + ETC
Proton Transport
Mito. Protein Degradation
Huntingtons Disease
Mito. Protein Import
Gen. of Precursor Metabolites
Protein Localization
Cristae Formation
Mito. Membrane Org.
Mito. Org.
AA Catabolism
Ketone Biosynthesis
FA Metabolism
Cell Differentiation
Cytoskel. Org.
Cell Cycle Checkpoints
Muscle Contraction
Positive Reg. Small GTPase Mediated Signal Transduction
Muscle Organ Development
Multicellular Organismal Movement
Myofibril Assembly
Striated Muscle Contraction
−2
−1
0
1
2
3
normalised enrichment score
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
100
200
300
400
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Training_LR, fgsea collapse survivors at FDR < 0.05

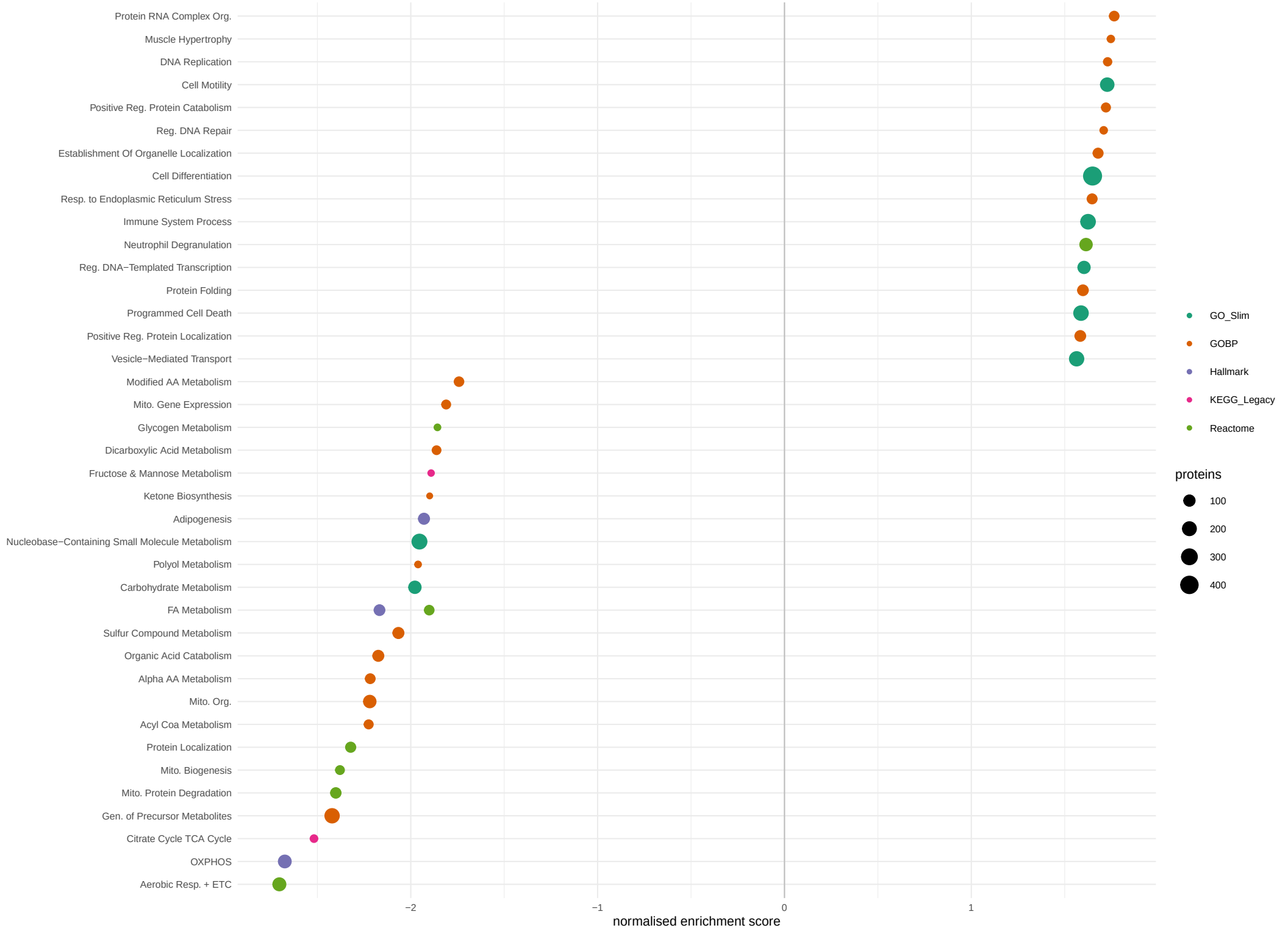
18 sets. Page 1 of 1.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Acute_HR, fgsea collapse survivors at FDR < 0.05

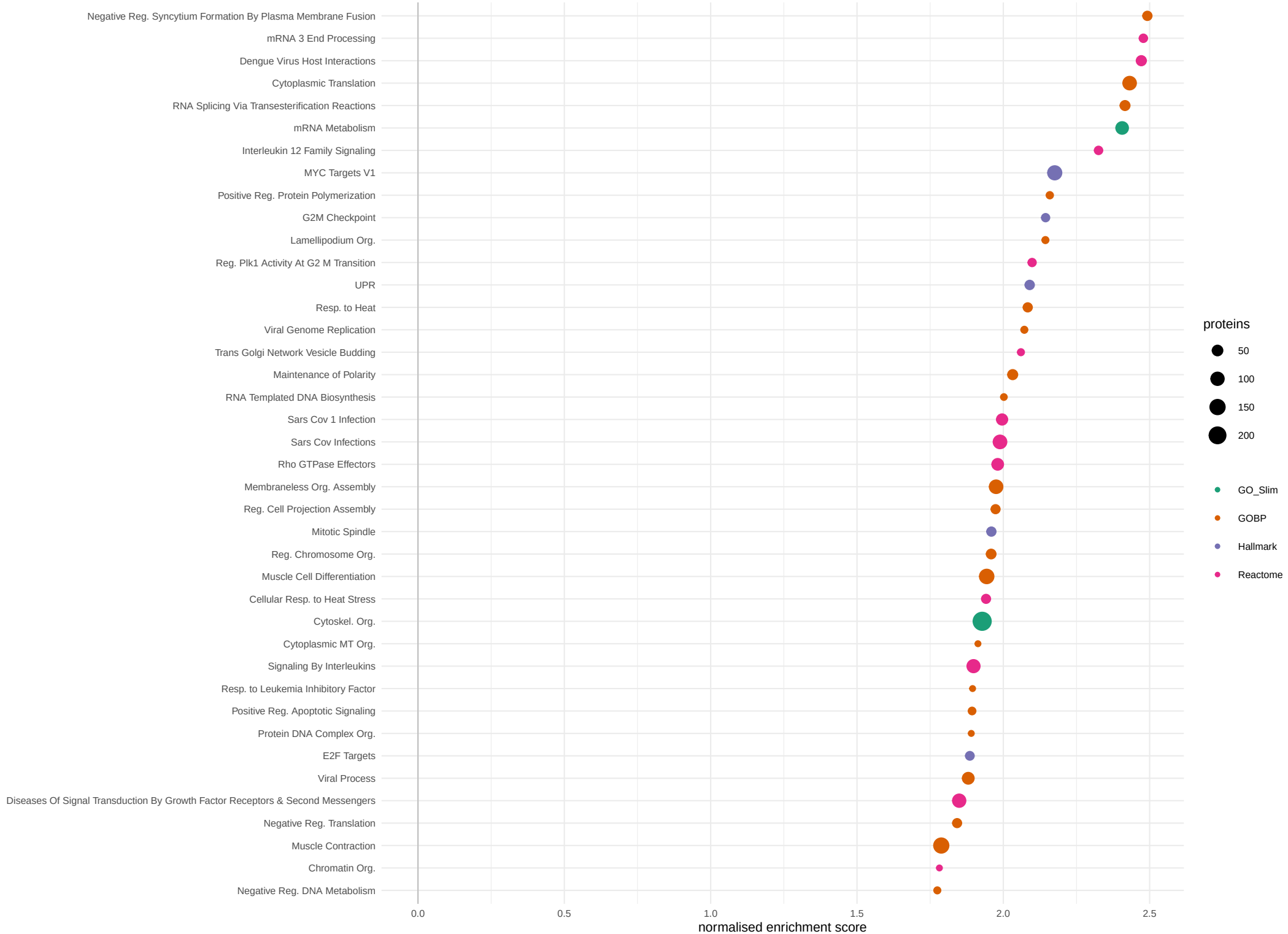
83 sets. Page 1 of 3.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Acute_HR, fgsea collapse survivors at FDR < 0.05

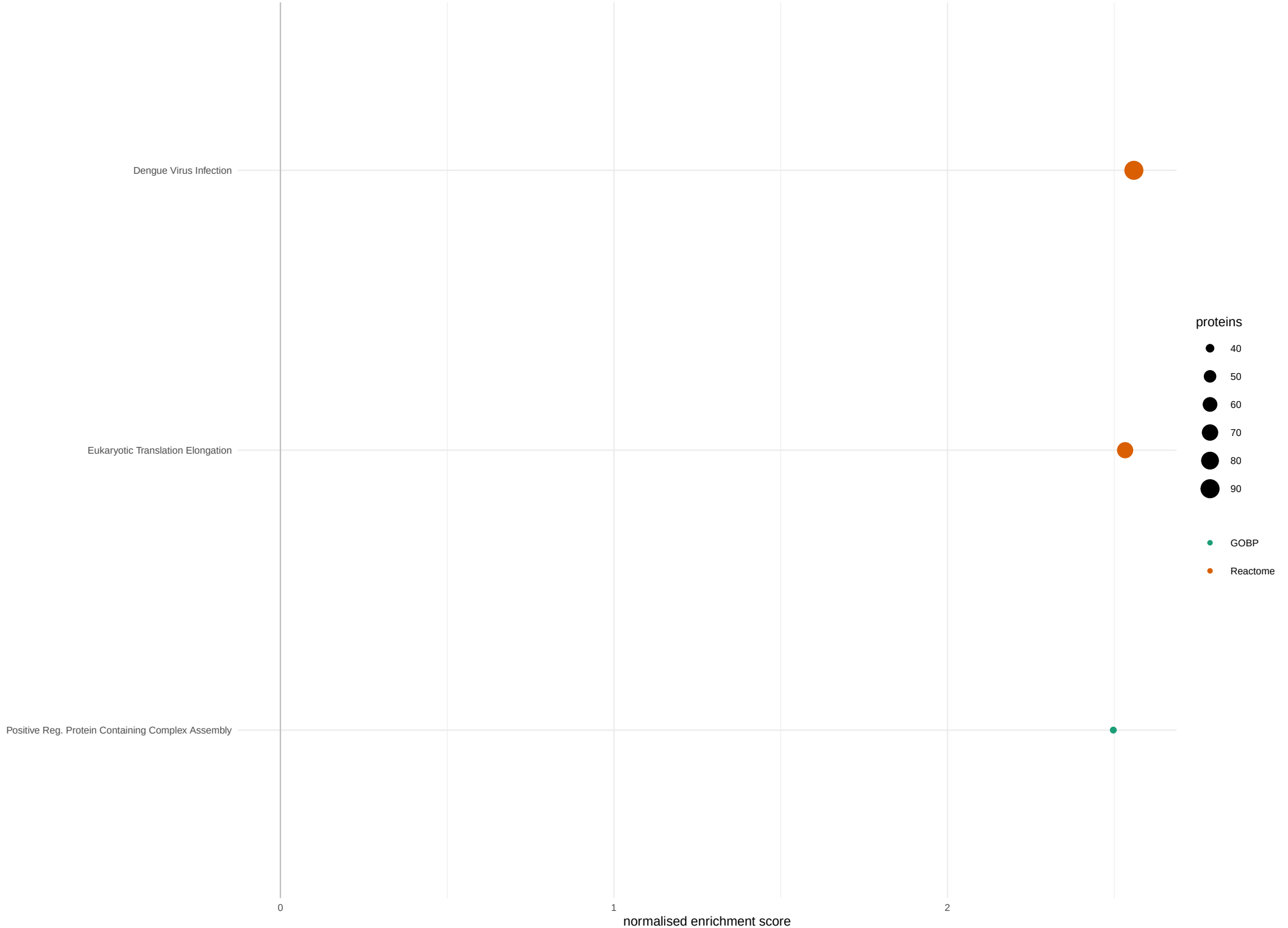
83 sets. Page 2 of 3.



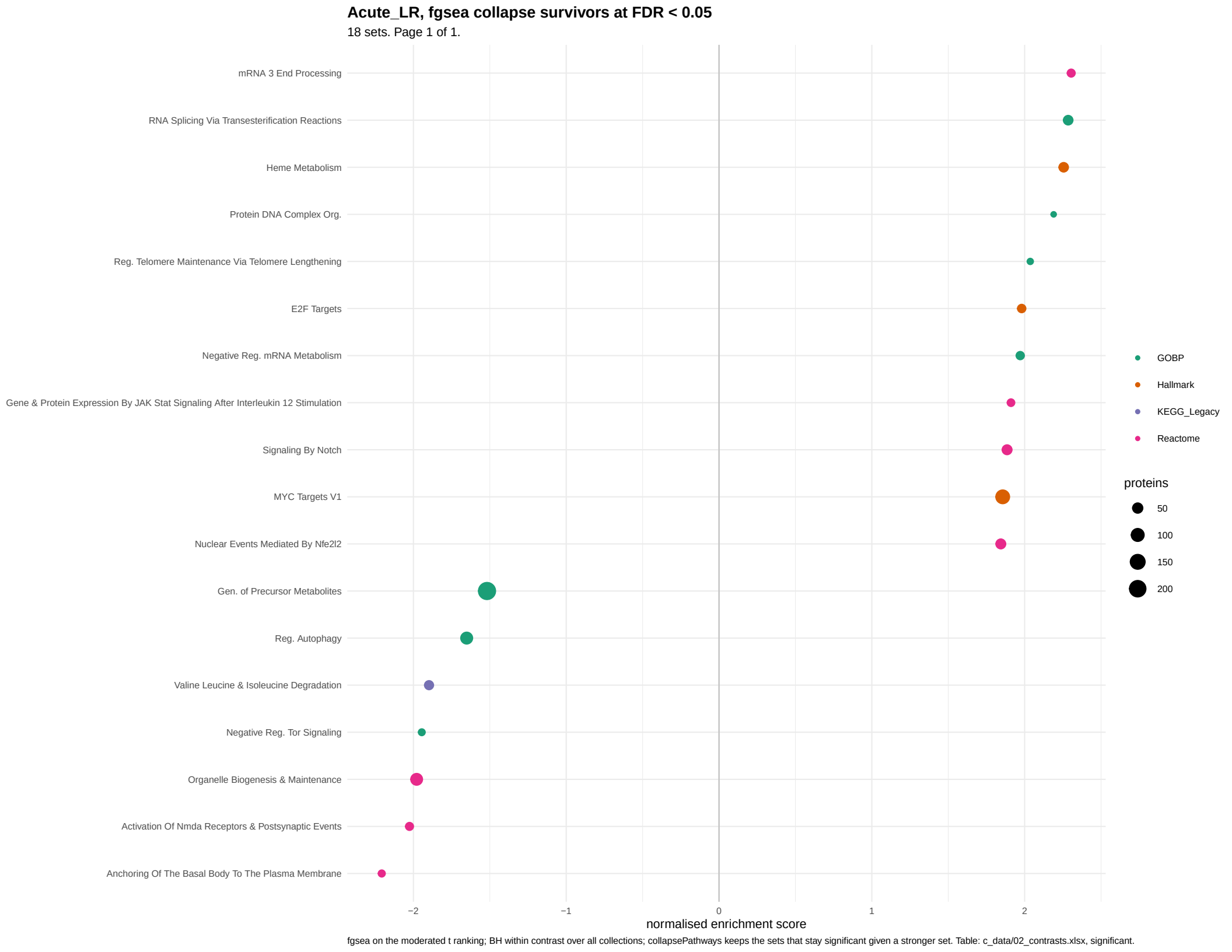
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Acute_HR, fgsea collapse survivors at FDR < 0.05

83 sets. Page 3 of 3.

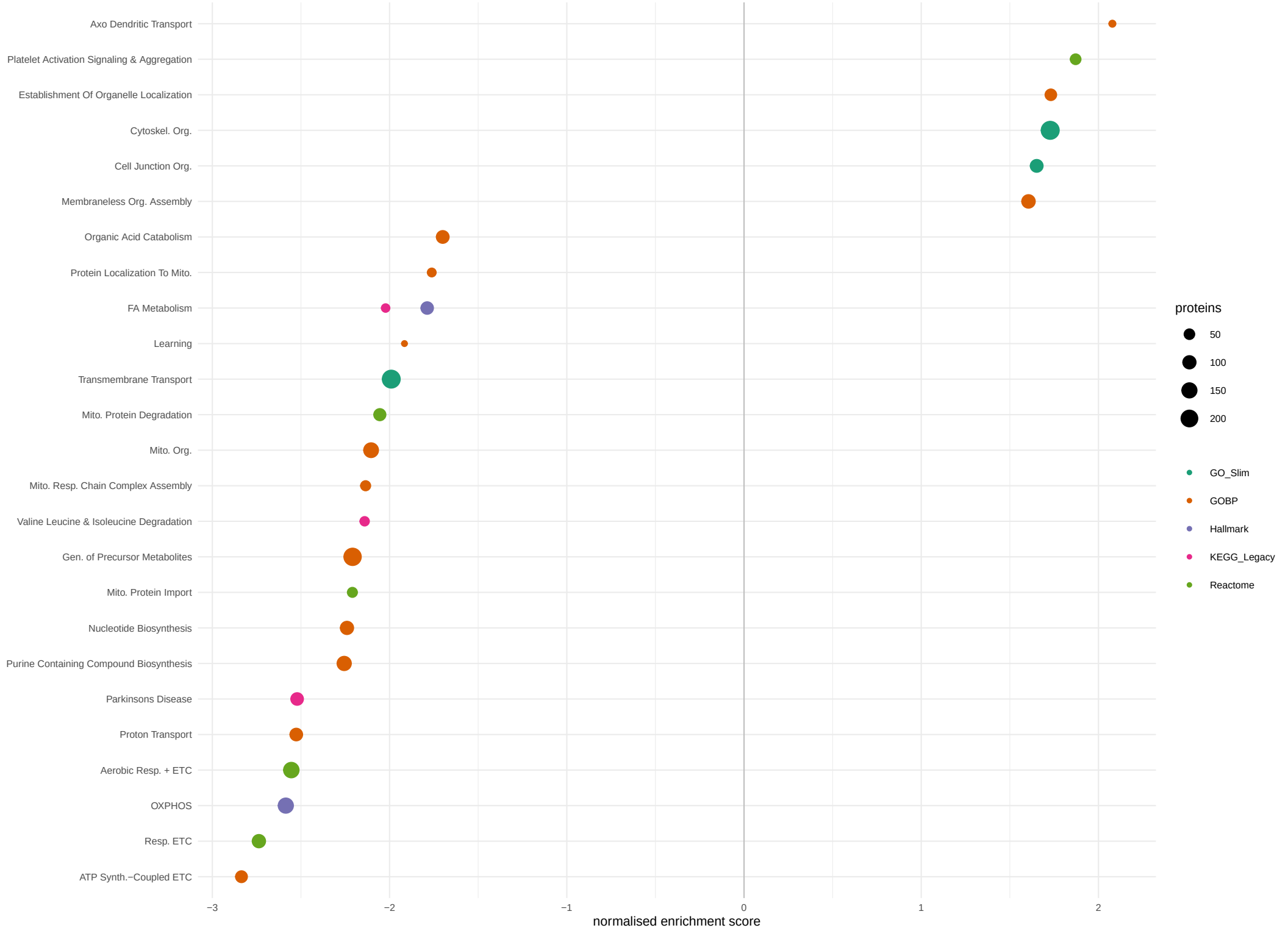


fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.


Acute_LR, fgsea collapse survivors at FDR < 0.05
18 sets. Page 1 of 1.
mRNA 3 End Processing
RNA Splicing Via Transesterification Reactions
Heme Metabolism
Protein DNA Complex Org.
Reg. Telomere Maintenance Via Telomere Lengthening
E2F Targets
Negative Reg. mRNA Metabolism
Gene & Protein Expression By JAK Stat Signaling After Interleukin 12 Stimulation
Signaling By Notch
MYC Targets V1
Nuclear Events Mediated By Nfe2l2
Gen. of Precursor Metabolites
Reg. Autophagy
Valine Leucine & Isoleucine Degradation
Negative Reg. Tor Signaling
Organelle Biogenesis & Maintenance
Activation Of Nmda Receptors & Postsynaptic Events
Anchoring Of The Basal Body To The Plasma Membrane
−2
−1
0
1
2
normalised enrichment score
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
150
200
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Baseline_HRvLR, fgsea collapse survivors at FDR < 0.05

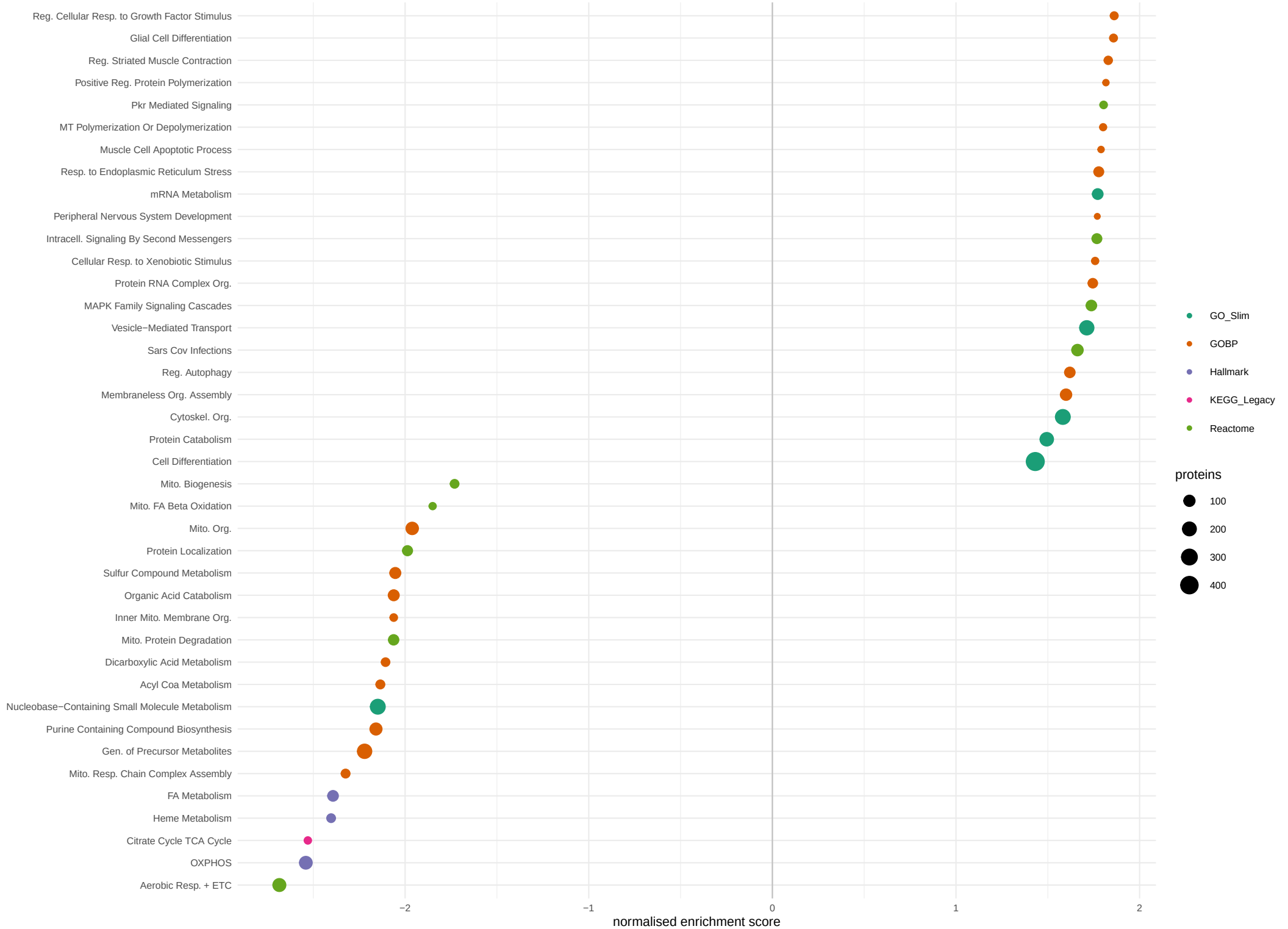
26 sets. Page 1 of 1.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Acute_HRvLR, fgsea collapse survivors at FDR < 0.05

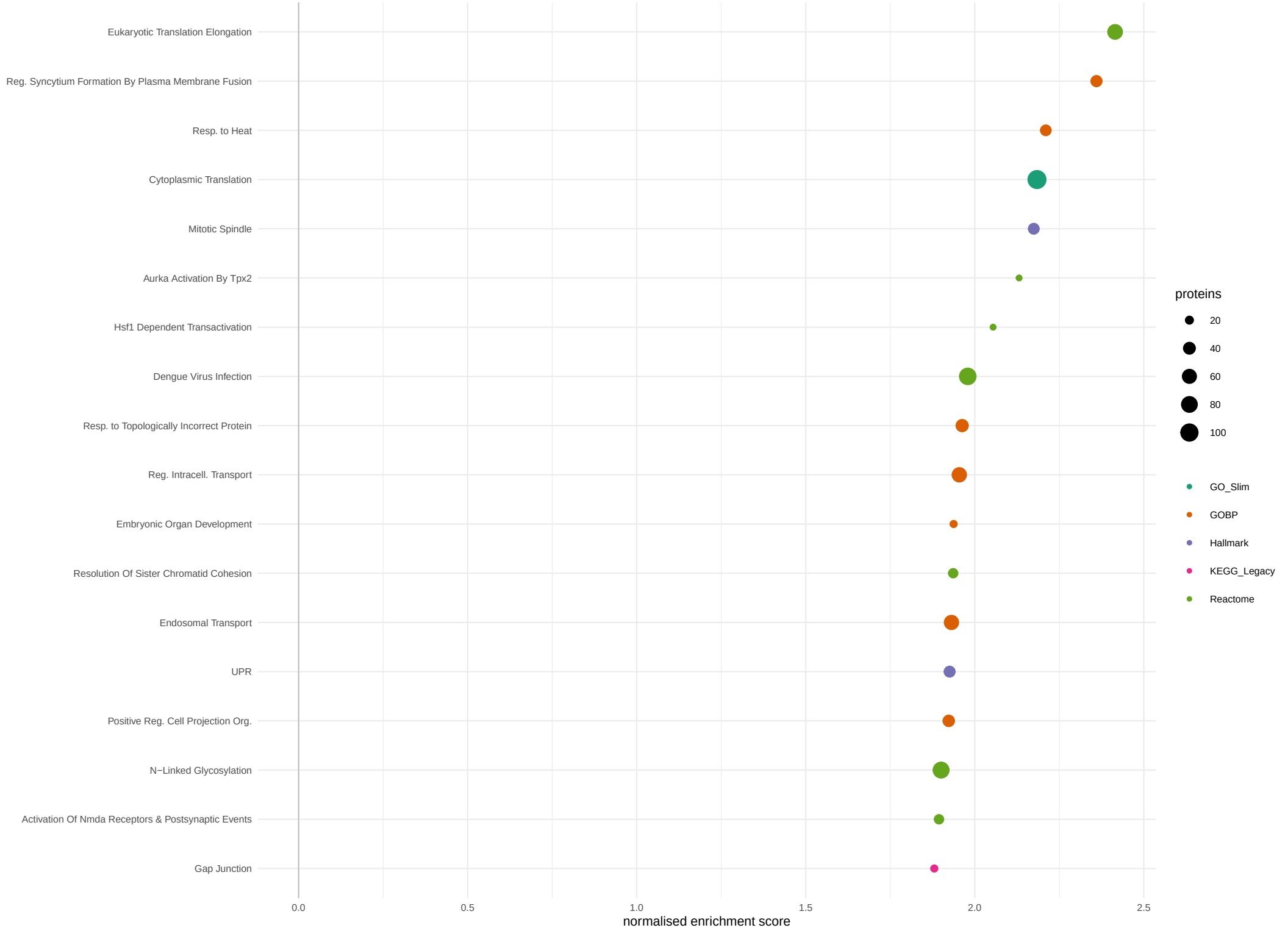
58 sets. Page 1 of 2.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Acute_HRvLR, fgsea collapse survivors at FDR < 0.05

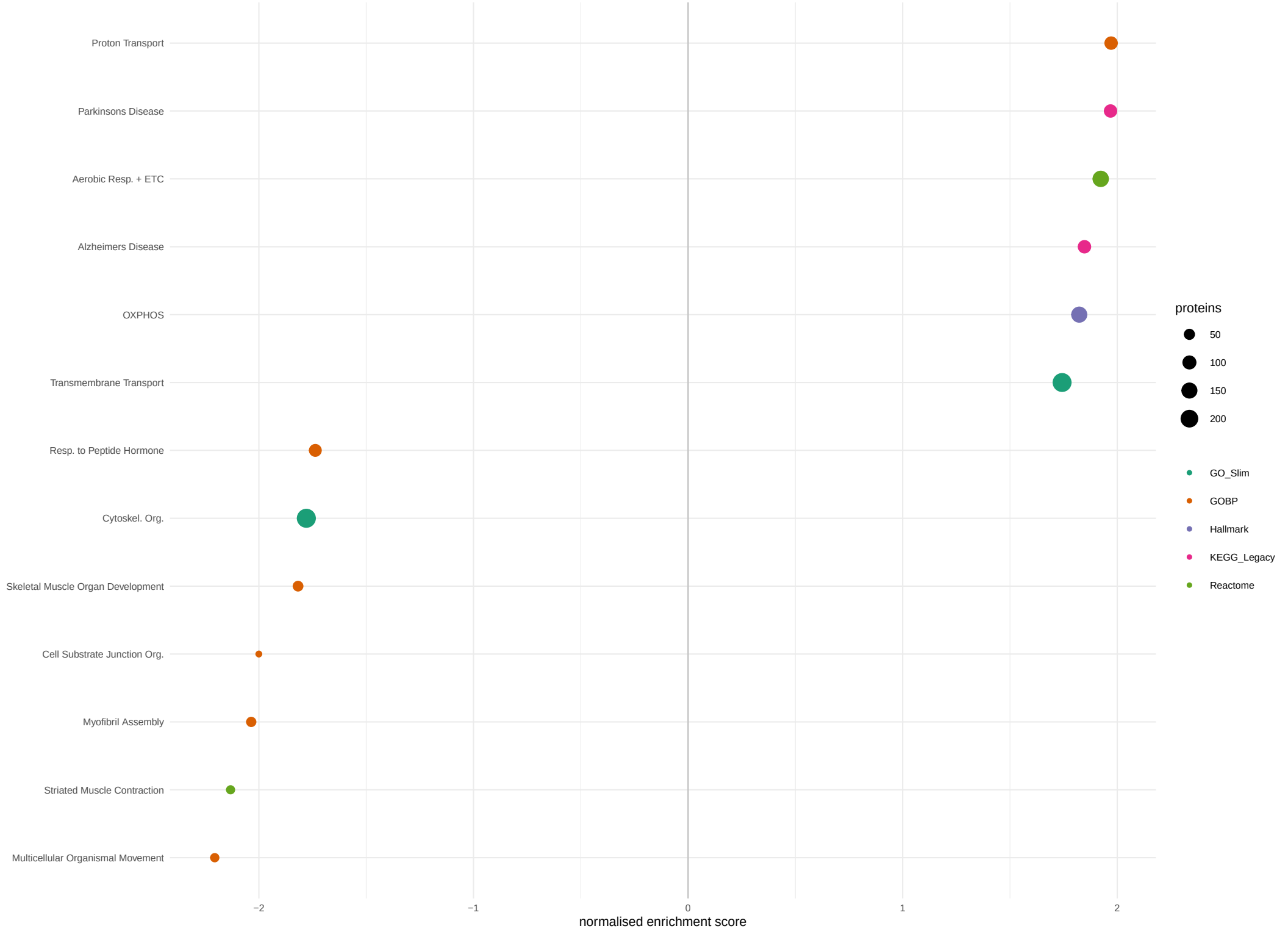
58 sets. Page 2 of 2.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Training_Interaction, fgsea collapse survivors at FDR < 0.05

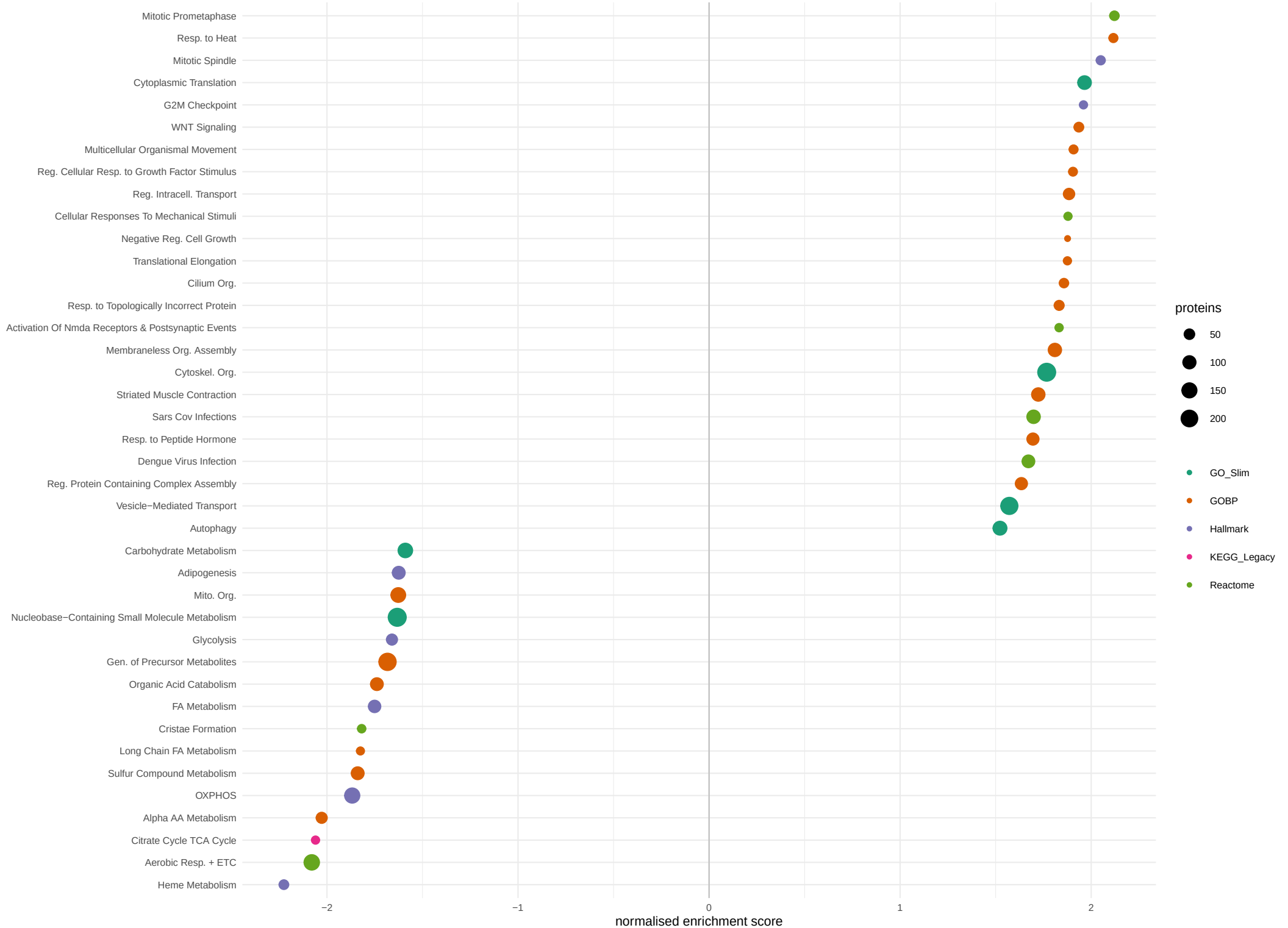
13 sets. Page 1 of 1.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Acute_Interaction, fgsea collapse survivors at FDR < 0.05

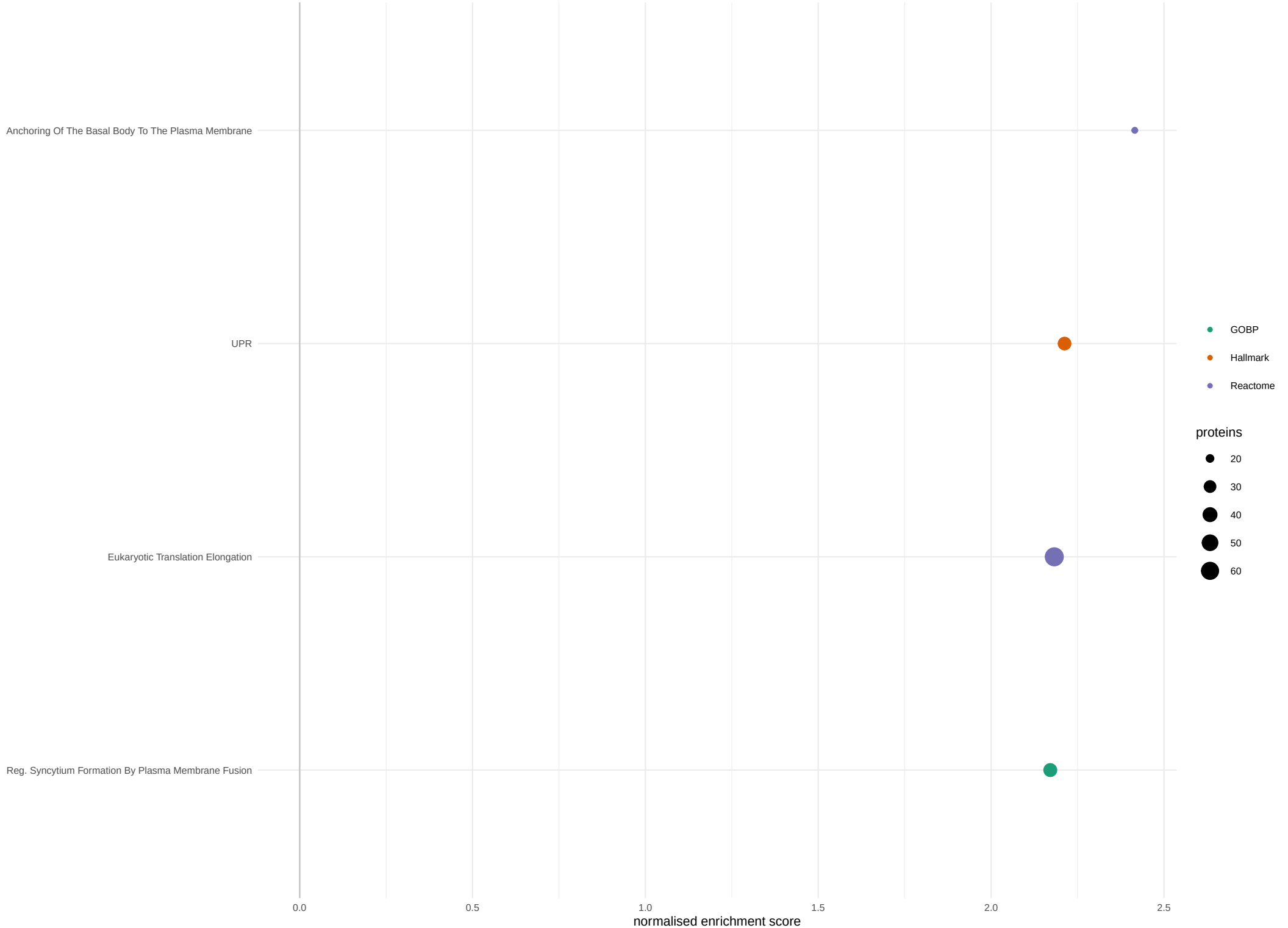
44 sets. Page 1 of 2.



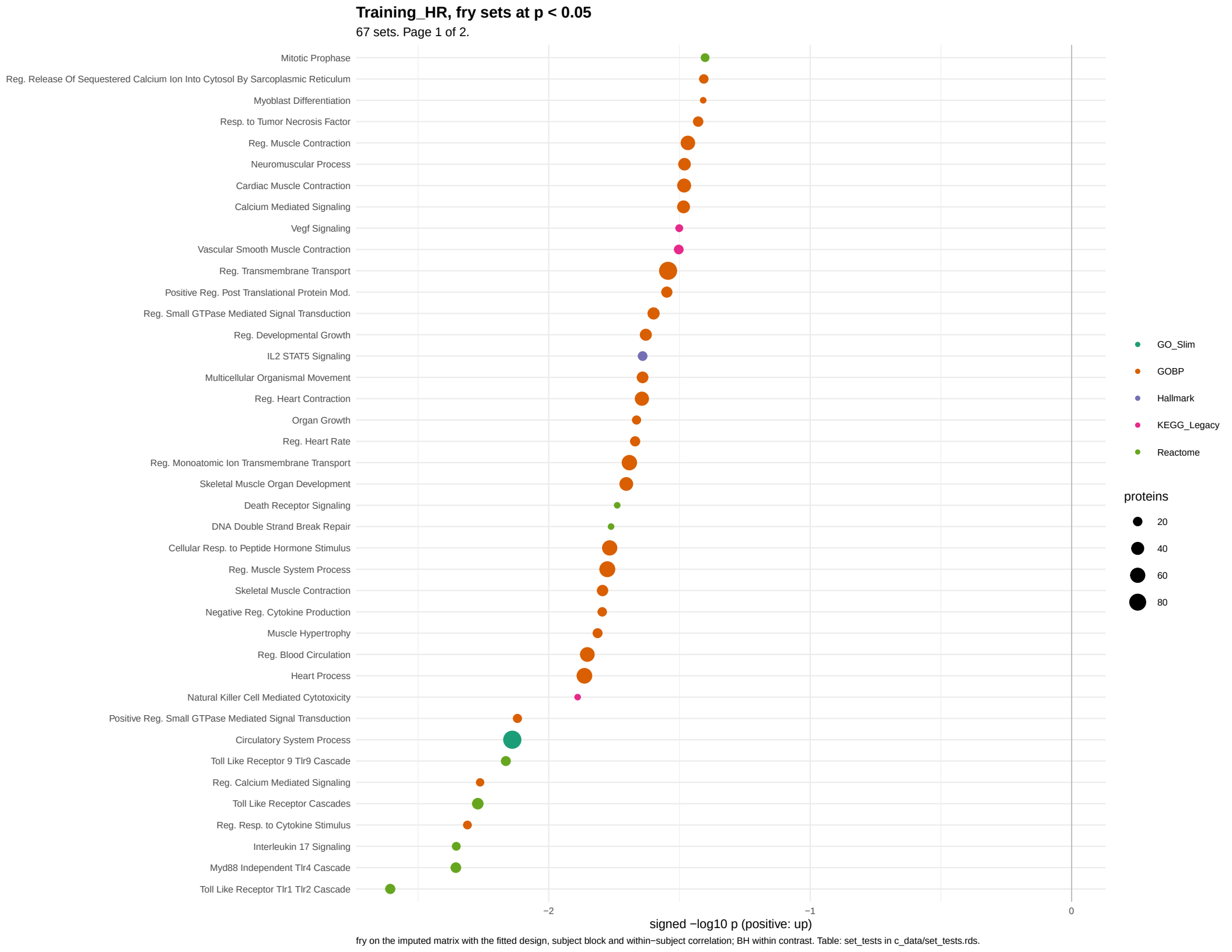
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Acute_Interaction, fgsea collapse survivors at FDR < 0.05

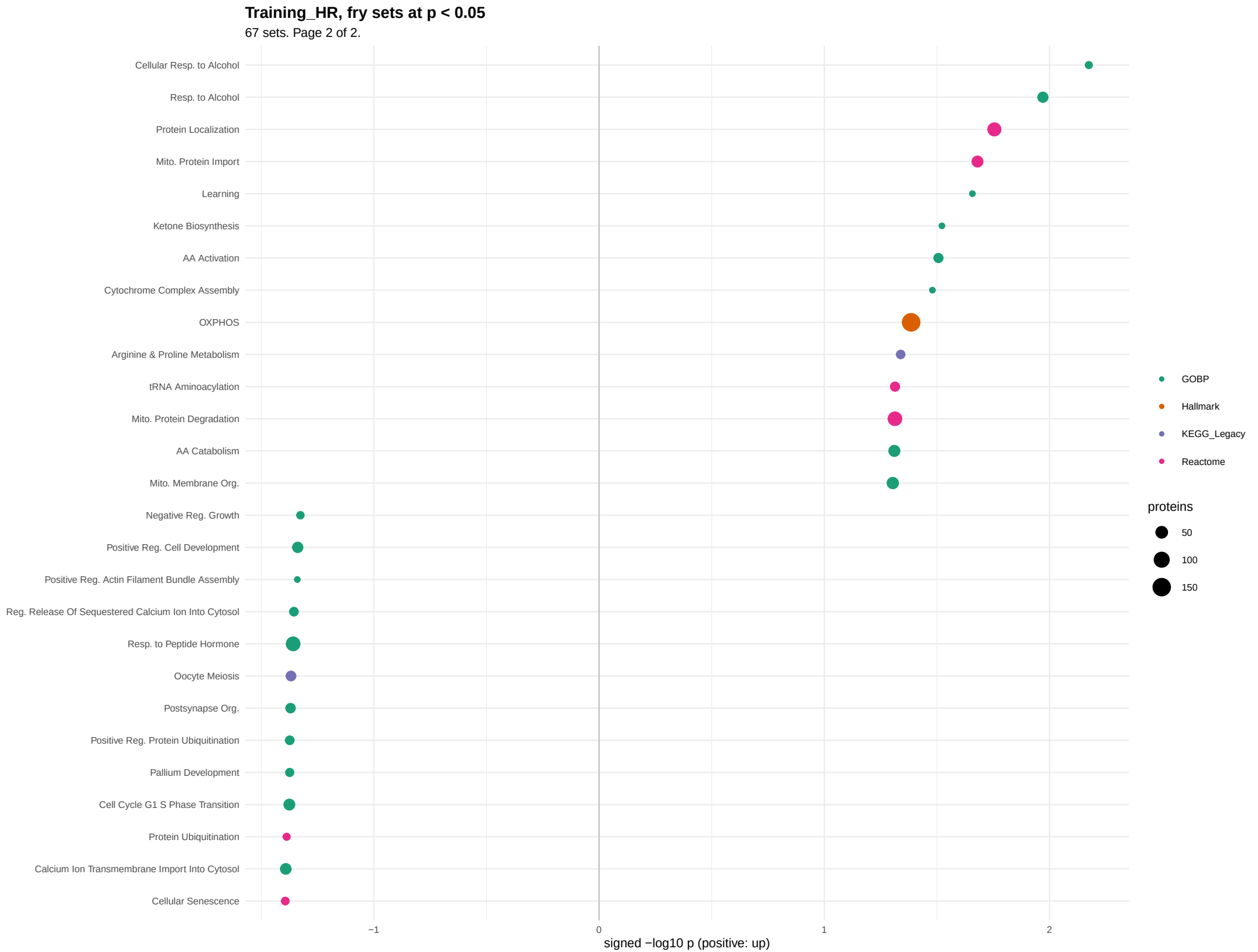
44 sets. Page 2 of 2.

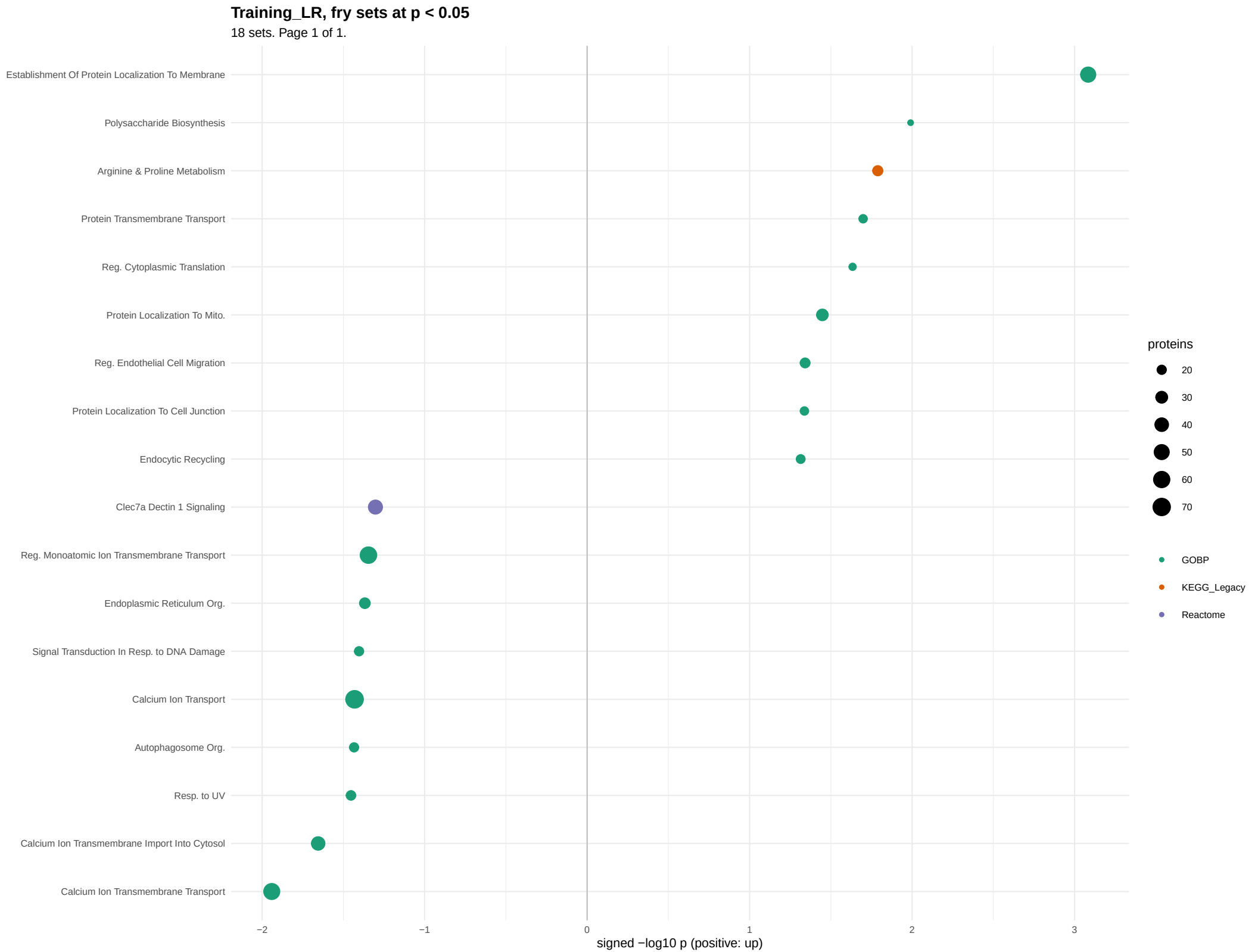


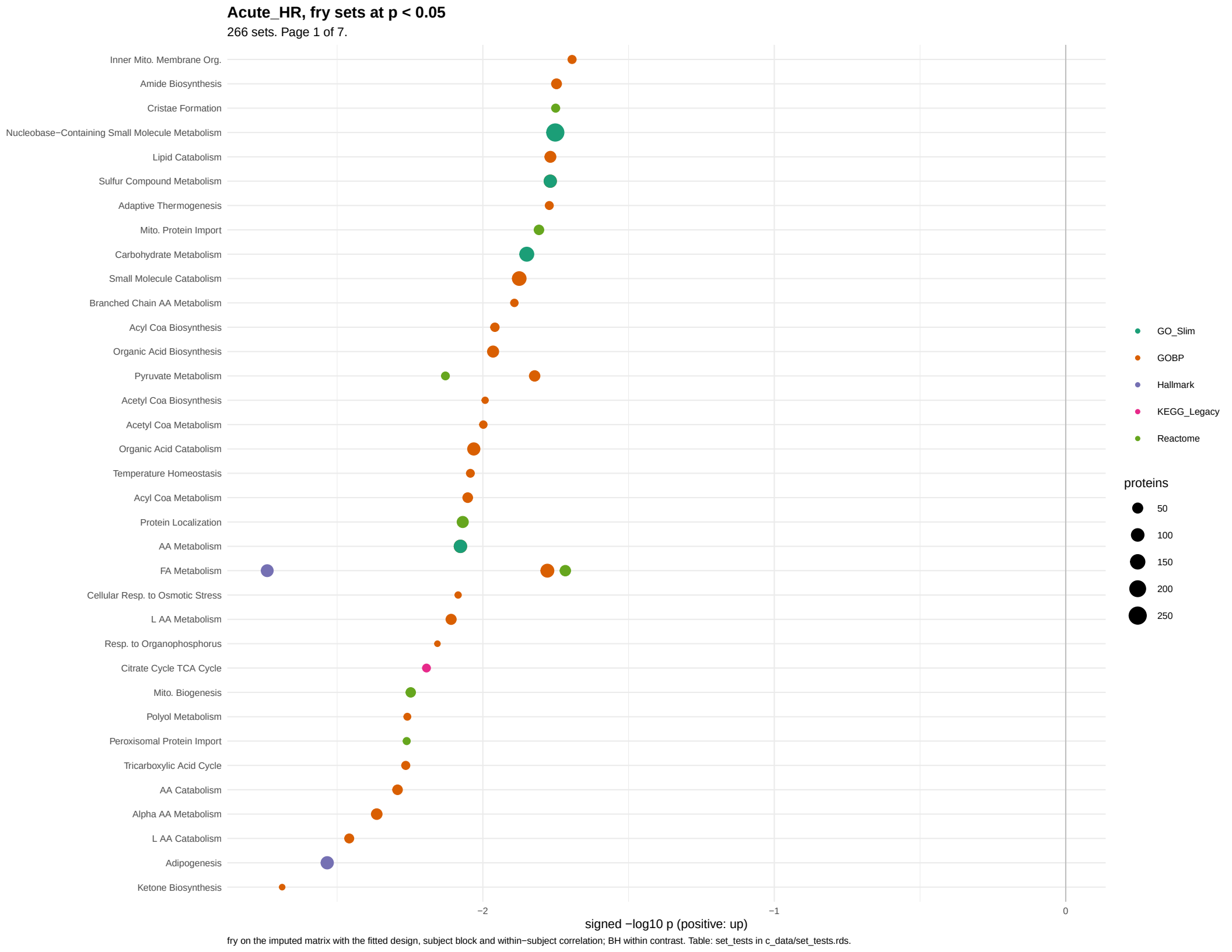
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

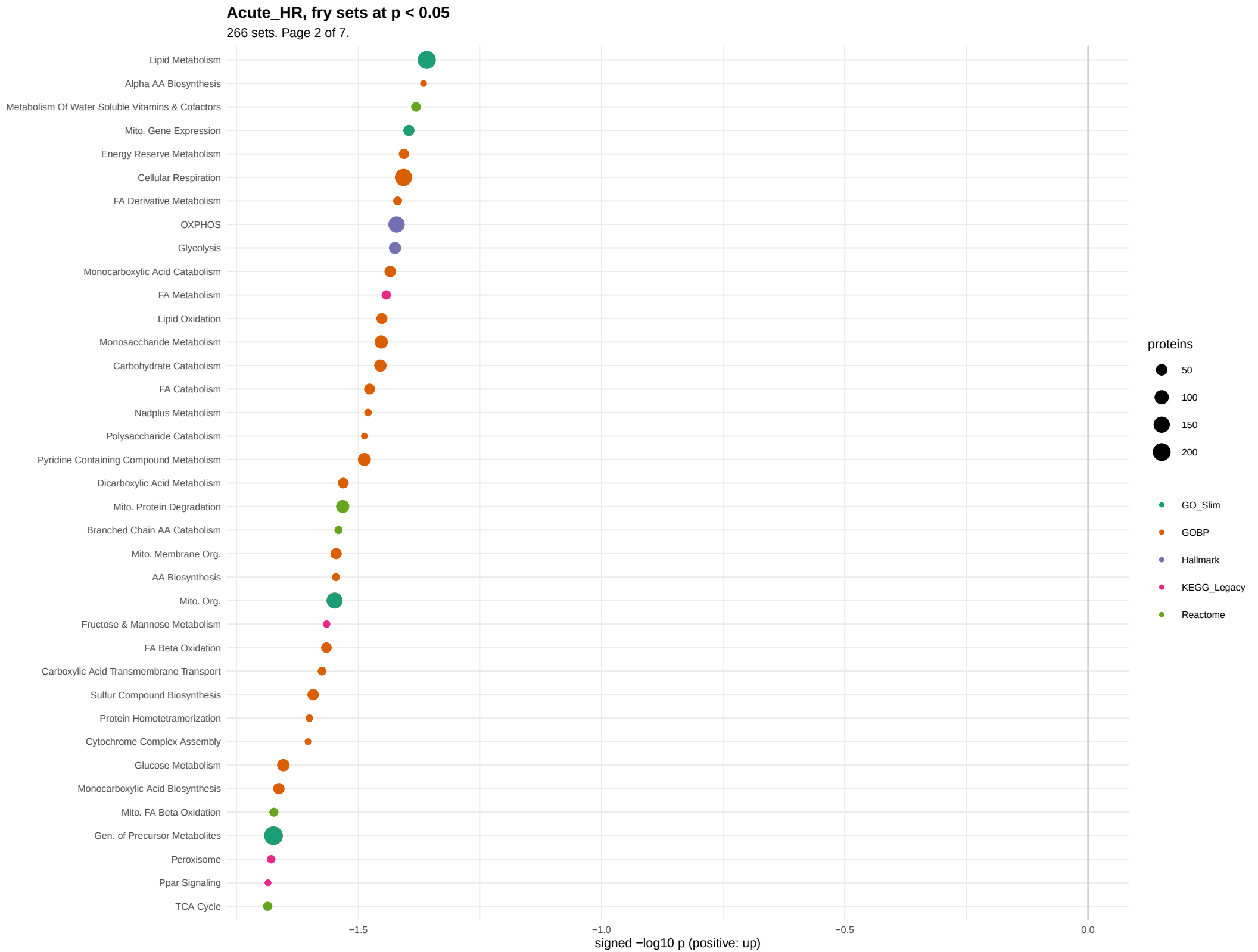


fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

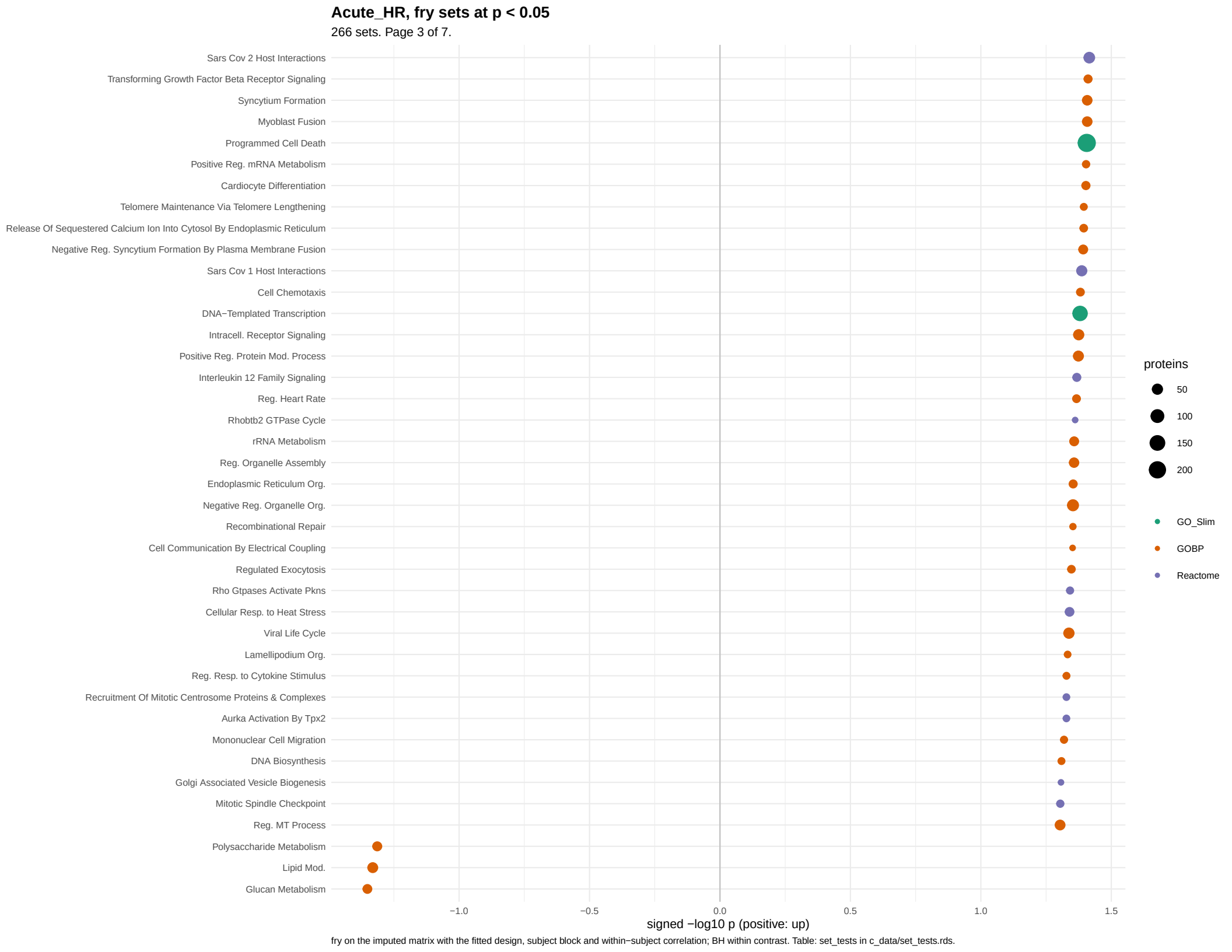

Training_HR, fry sets at p < 0.05
67 sets. Page 2 of 2.
Cellular Resp. to Alcohol
Resp. to Alcohol
Protein Localization
Mito. Protein Import
Learning
Ketone Biosynthesis
AA Activation
Cytochrome Complex Assembly
OXPHOS
Arginine & Proline Metabolism
tRNA Aminoacylation
Mito. Protein Degradation
AA Catabolism
Mito. Membrane Org.
Negative Reg. Growth
Positive Reg. Cell Development
Positive Reg. Actin Filament Bundle Assembly
Reg. Release Of Sequestered Calcium Ion Into Cytosol
Resp. to Peptide Hormone
Oocyte Meiosis
Postsynapse Org.
Positive Reg. Protein Ubiquitination
Pallium Development
Cell Cycle G1 S Phase Transition
Protein Ubiquitination
Calcium Ion Transmembrane Import Into Cytosol
Cellular Senescence
−1
0
1
2
signed −log10 p (positive: up)
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
150
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

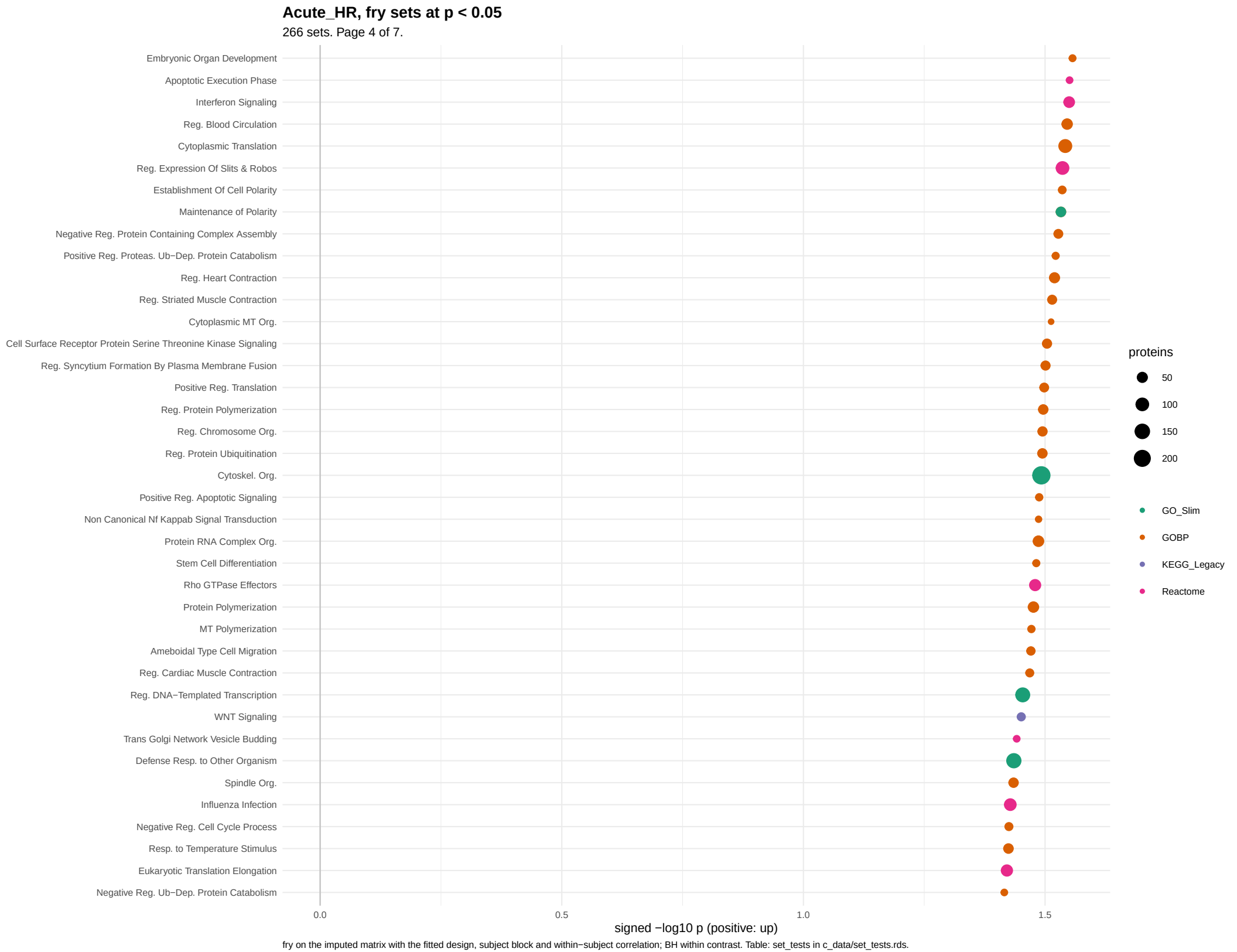

Training_LR, fry sets at p < 0.05
18 sets. Page 1 of 1.
Establishment Of Protein Localization To Membrane
Polysaccharide Biosynthesis
Arginine & Proline Metabolism
Protein Transmembrane Transport
Reg. Cytoplasmic Translation
Protein Localization To Mito.
Reg. Endothelial Cell Migration
Protein Localization To Cell Junction
Endocytic Recycling
Clec7a Dectin 1 Signaling
Reg. Monoatomic Ion Transmembrane Transport
Endoplasmic Reticulum Org.
Signal Transduction In Resp. to DNA Damage
Calcium Ion Transport
Autophagosome Org.
Resp. to UV
Calcium Ion Transmembrane Import Into Cytosol
Calcium Ion Transmembrane Transport
−2
−1
0
1
2
3
signed −log10 p (positive: up)
proteins
20
30
40
50
60
70
GOBP
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_HR, fry sets at p < 0.05
266 sets. Page 1 of 7.
Inner Mito. Membrane Org.
Amide Biosynthesis
Cristae Formation
Nucleobase−Containing Small Molecule Metabolism
Lipid Catabolism
Sulfur Compound Metabolism
Adaptive Thermogenesis
Mito. Protein Import
Carbohydrate Metabolism
Small Molecule Catabolism
Branched Chain AA Metabolism
Acyl Coa Biosynthesis
Organic Acid Biosynthesis
Pyruvate Metabolism
Acetyl Coa Biosynthesis
Acetyl Coa Metabolism
Organic Acid Catabolism
Temperature Homeostasis
Acyl Coa Metabolism
Protein Localization
AA Metabolism
FA Metabolism
Cellular Resp. to Osmotic Stress
L AA Metabolism
Resp. to Organophosphorus
Citrate Cycle TCA Cycle
Mito. Biogenesis
Polyol Metabolism
Peroxisomal Protein Import
Tricarboxylic Acid Cycle
AA Catabolism
Alpha AA Metabolism
L AA Catabolism
Adipogenesis
Ketone Biosynthesis
−2
−1
0
signed −log10 p (positive: up)
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
150
200
250
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.



fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_HR, fry sets at p < 0.05
266 sets. Page 3 of 7.
Sars Cov 2 Host Interactions
Transforming Growth Factor Beta Receptor Signaling
Syncytium Formation
Myoblast Fusion
Programmed Cell Death
Positive Reg. mRNA Metabolism
Cardiocyte Differentiation
Telomere Maintenance Via Telomere Lengthening
Release Of Sequestered Calcium Ion Into Cytosol By Endoplasmic Reticulum
Negative Reg. Syncytium Formation By Plasma Membrane Fusion
Sars Cov 1 Host Interactions
Cell Chemotaxis
DNA−Templated Transcription
Intracell. Receptor Signaling
Positive Reg. Protein Mod. Process
Interleukin 12 Family Signaling
Reg. Heart Rate
Rhobtb2 GTPase Cycle
rRNA Metabolism
Reg. Organelle Assembly
Endoplasmic Reticulum Org.
Negative Reg. Organelle Org.
Recombinational Repair
Cell Communication By Electrical Coupling
Regulated Exocytosis
Rho Gtpases Activate Pkns
Cellular Resp. to Heat Stress
Viral Life Cycle
Lamellipodium Org.
Reg. Resp. to Cytokine Stimulus
Recruitment Of Mitotic Centrosome Proteins & Complexes
Aurka Activation By Tpx2
Mononuclear Cell Migration
DNA Biosynthesis
Golgi Associated Vesicle Biogenesis
Mitotic Spindle Checkpoint
Reg. MT Process
Polysaccharide Metabolism
Lipid Mod.
Glucan Metabolism
−1.0
−0.5
0.0
0.5
1.0
1.5
signed −log10 p (positive: up)
proteins
50
100
150
200
GO_Slim
GOBP
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_HR, fry sets at p < 0.05
266 sets. Page 4 of 7.
Embryonic Organ Development
Apoptotic Execution Phase
Interferon Signaling
Reg. Blood Circulation
Cytoplasmic Translation
Reg. Expression Of Slits & Robos
Establishment Of Cell Polarity
Maintenance of Polarity
Negative Reg. Protein Containing Complex Assembly
Positive Reg. Proteas. Ub−Dep. Protein Catabolism
Reg. Heart Contraction
Reg. Striated Muscle Contraction
Cytoplasmic MT Org.
Cell Surface Receptor Protein Serine Threonine Kinase Signaling
Reg. Syncytium Formation By Plasma Membrane Fusion
Positive Reg. Translation
Reg. Protein Polymerization
Reg. Chromosome Org.
Reg. Protein Ubiquitination
Cytoskel. Org.
Positive Reg. Apoptotic Signaling
Non Canonical Nf Kappab Signal Transduction
Protein RNA Complex Org.
Stem Cell Differentiation
Rho GTPase Effectors
Protein Polymerization
MT Polymerization
Ameboidal Type Cell Migration
Reg. Cardiac Muscle Contraction
Reg. DNA−Templated Transcription
WNT Signaling
Trans Golgi Network Vesicle Budding
Defense Resp. to Other Organism
Spindle Org.
Influenza Infection
Negative Reg. Cell Cycle Process
Resp. to Temperature Stimulus
Eukaryotic Translation Elongation
Negative Reg. Ub−Dep. Protein Catabolism
0.0
0.5
1.0
1.5
signed −log10 p (positive: up)
proteins
50
100
150
200
GO_Slim
GOBP
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

Acute_HR, fry sets at p < 0.05

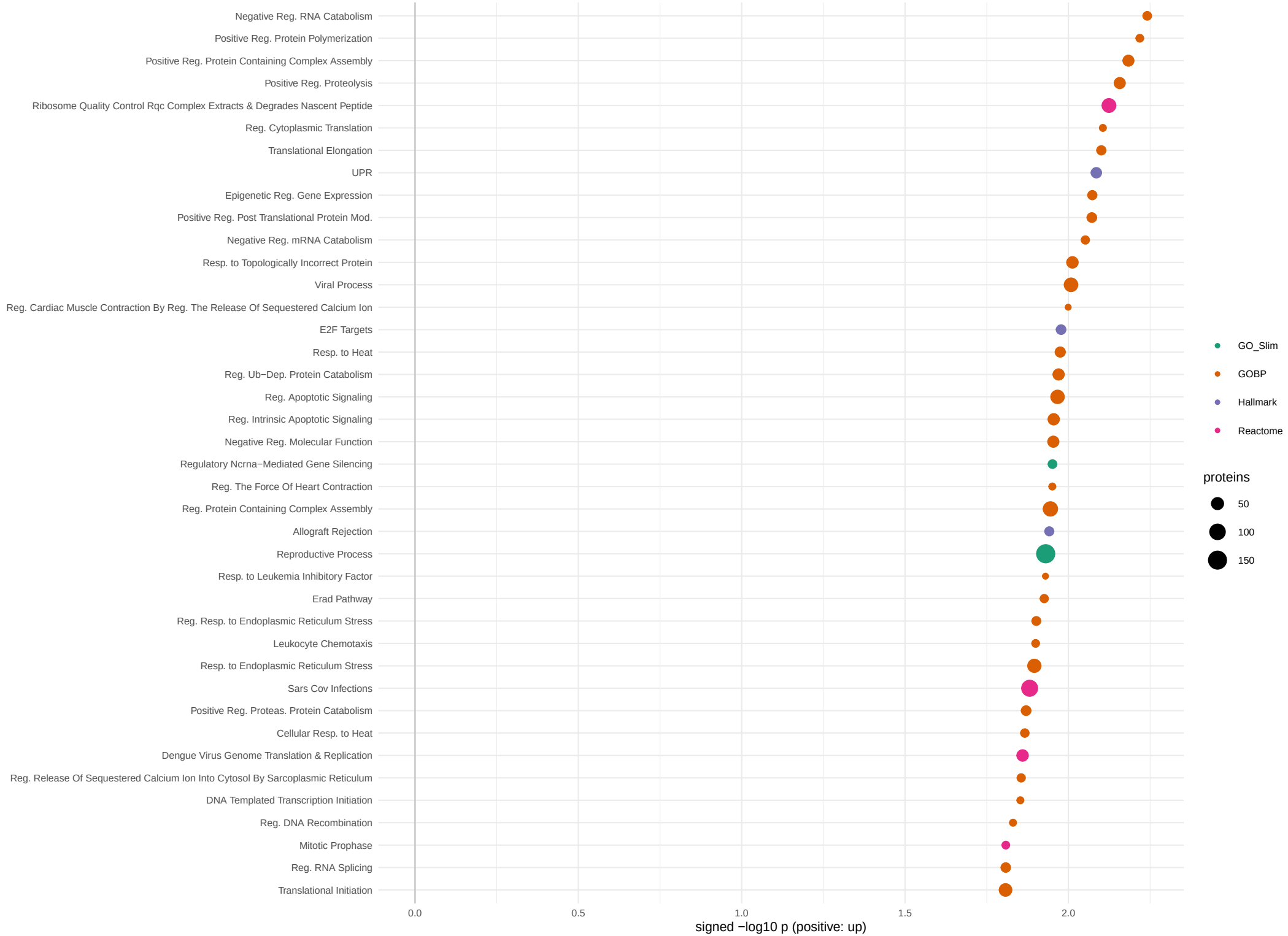
266 sets. Page 5 of 7.



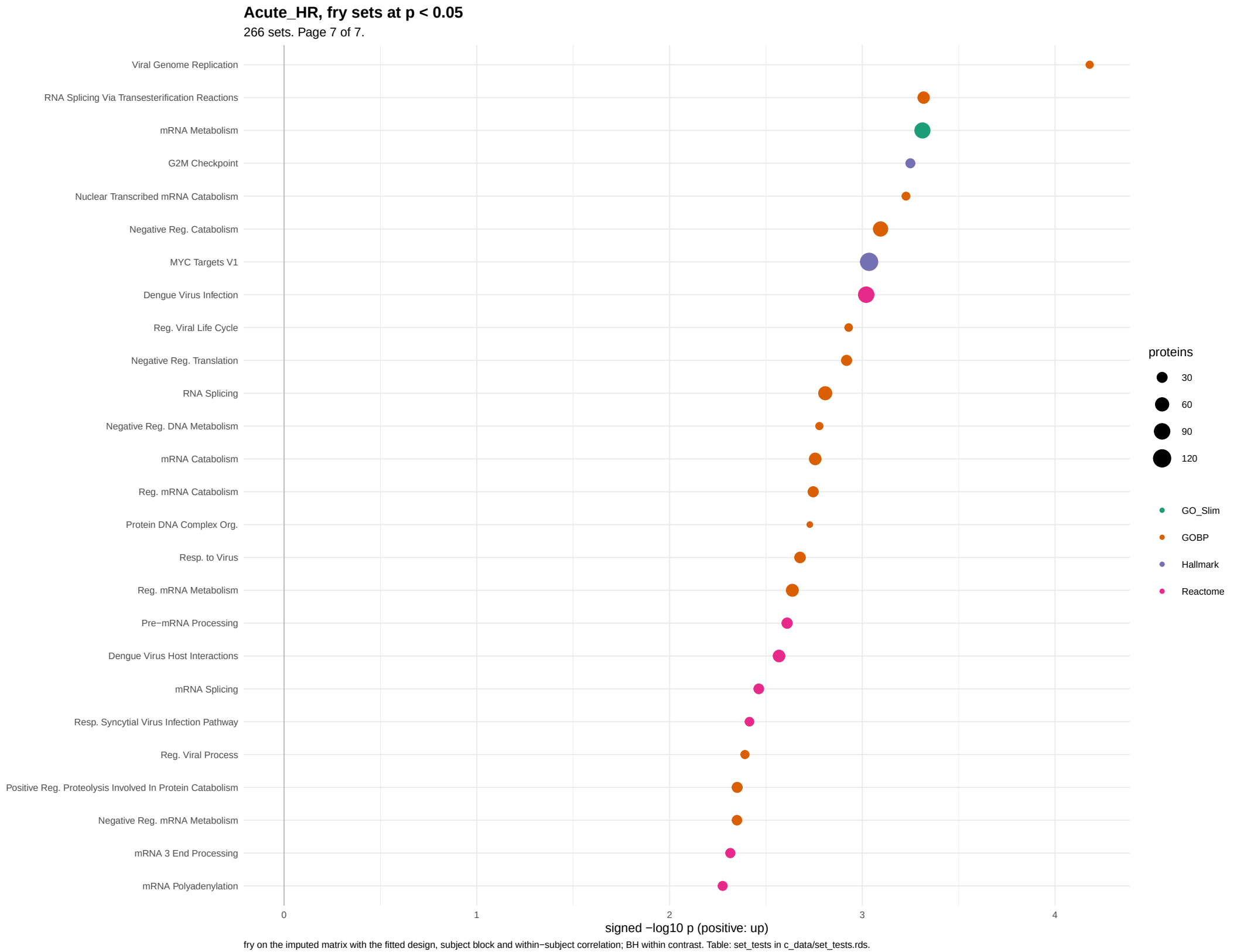
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

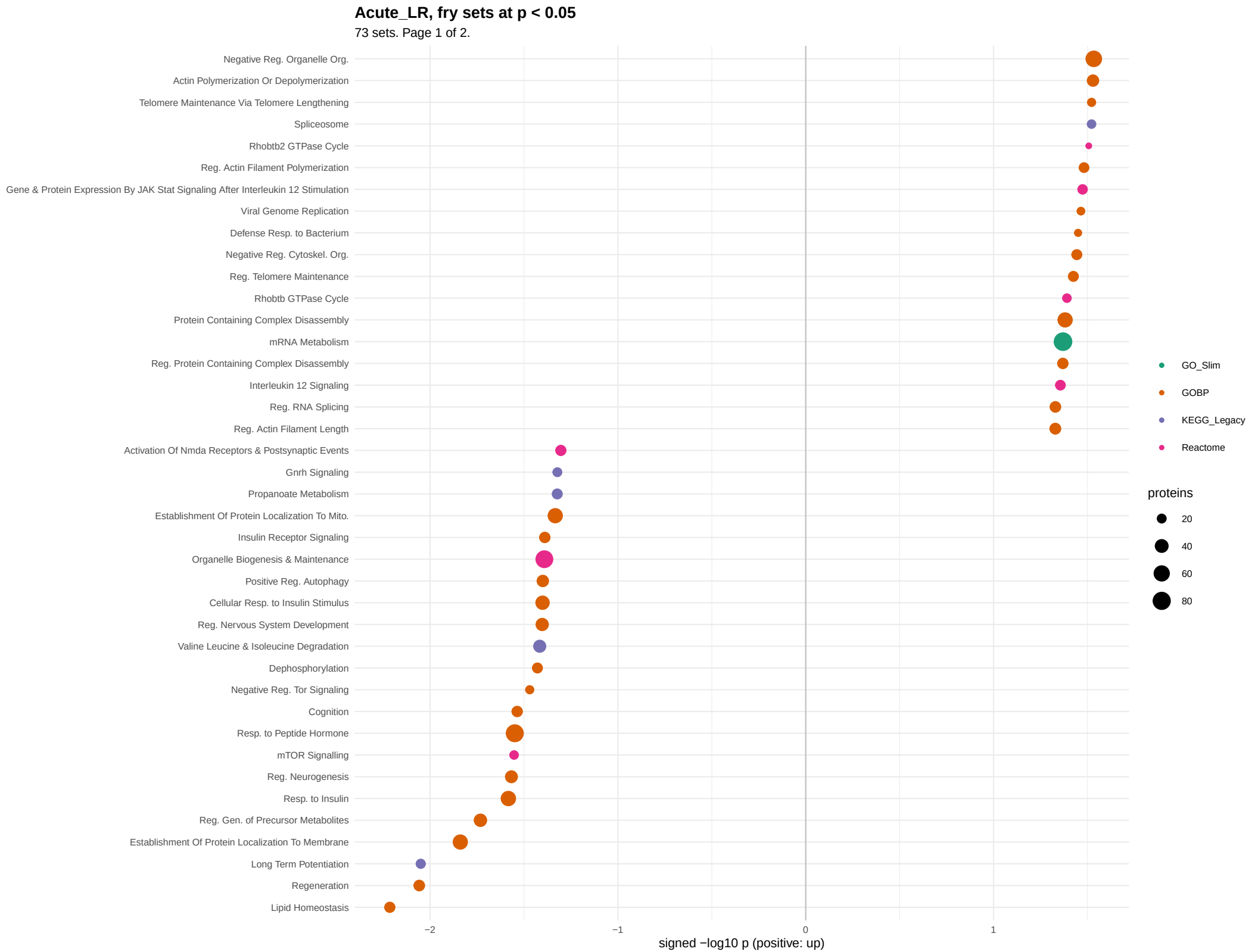
Acute_HR, fry sets at p < 0.05

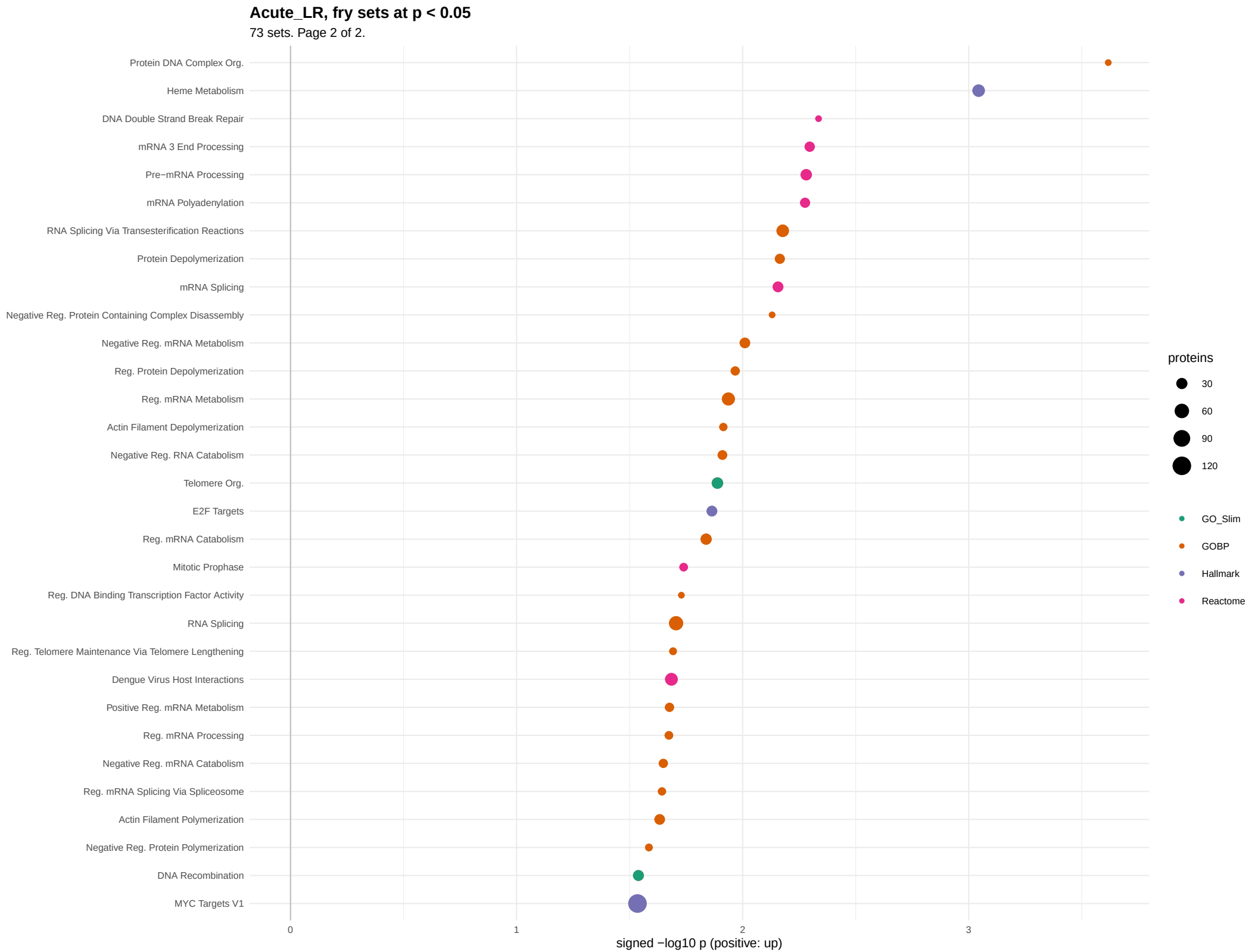
266 sets. Page 6 of 7.

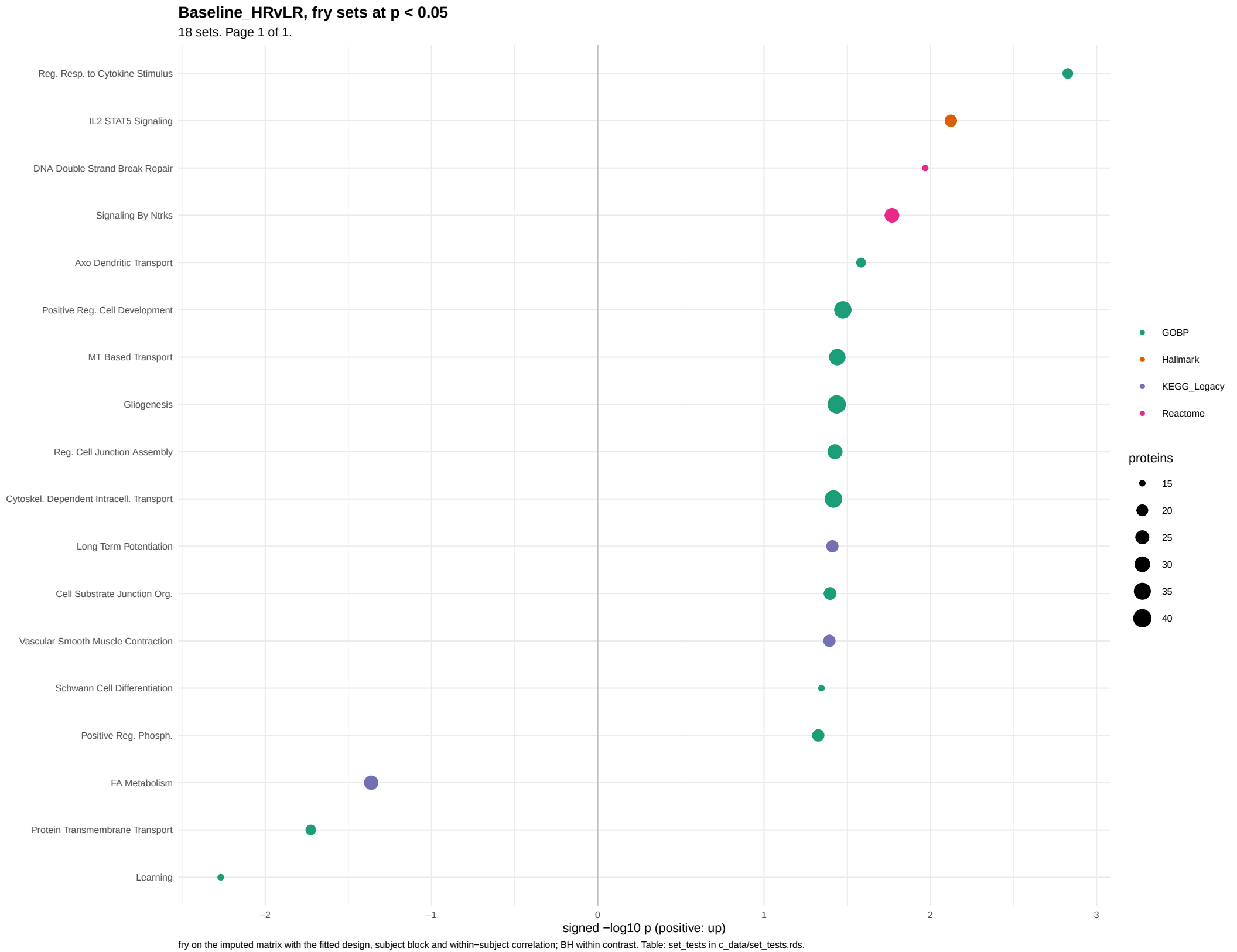


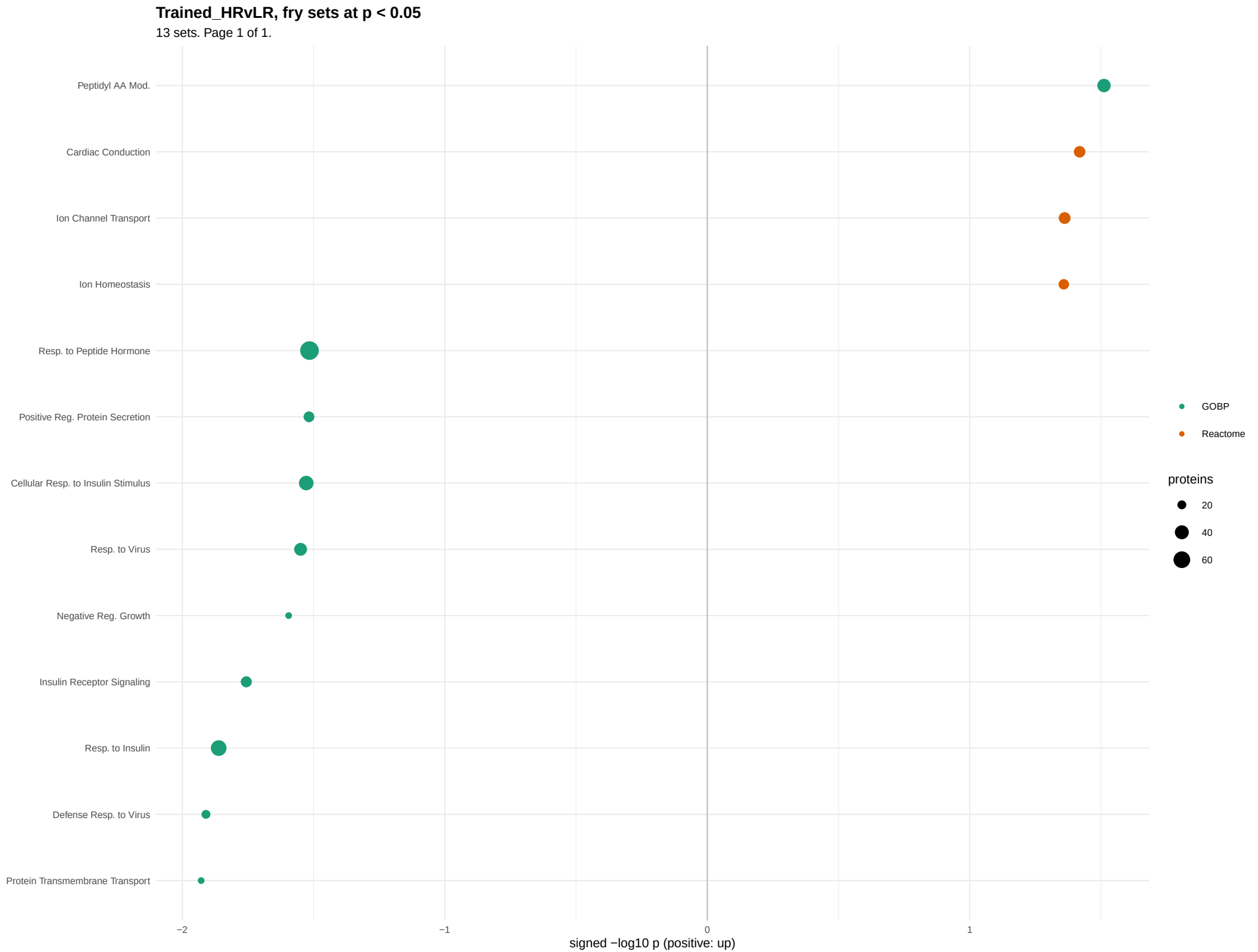
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

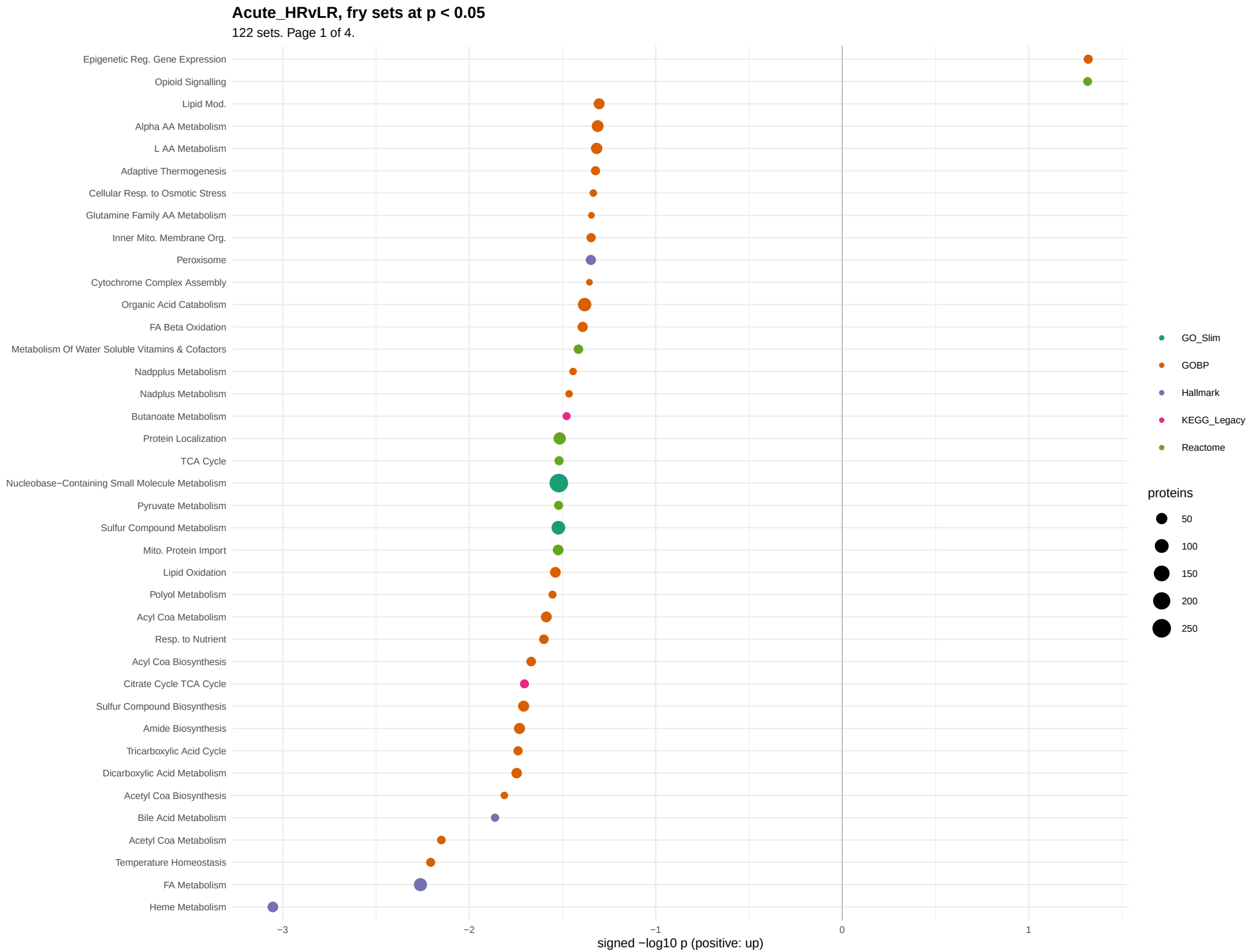

Acute_HR, fry sets at p < 0.05
266 sets. Page 7 of 7.
Viral Genome Replication
RNA Splicing Via Transesterification Reactions
mRNA Metabolism
G2M Checkpoint
Nuclear Transcribed mRNA Catabolism
Negative Reg. Catabolism
MYC Targets V1
Dengue Virus Infection
Reg. Viral Life Cycle
Negative Reg. Translation
RNA Splicing
Negative Reg. DNA Metabolism
mRNA Catabolism
Reg. mRNA Catabolism
Protein DNA Complex Org.
Resp. to Virus
Reg. mRNA Metabolism
Pre−mRNA Processing
Dengue Virus Host Interactions
mRNA Splicing
Resp. Syncytial Virus Infection Pathway
Reg. Viral Process
Positive Reg. Proteolysis Involved In Protein Catabolism
Negative Reg. mRNA Metabolism
mRNA 3 End Processing
mRNA Polyadenylation
0
1
2
3
4
signed −log10 p (positive: up)
proteins
30
60
90
120
GO_Slim
GOBP
Hallmark
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_LR, fry sets at p < 0.05
73 sets. Page 1 of 2.
Negative Reg. Organelle Org.
Actin Polymerization Or Depolymerization
Telomere Maintenance Via Telomere Lengthening
Spliceosome
Rhobtb2 GTPase Cycle
Reg. Actin Filament Polymerization
Gene & Protein Expression By JAK Stat Signaling After Interleukin 12 Stimulation
Viral Genome Replication
Defense Resp. to Bacterium
Negative Reg. Cytoskel. Org.
Reg. Telomere Maintenance
Rhobtb GTPase Cycle
Protein Containing Complex Disassembly
mRNA Metabolism
Reg. Protein Containing Complex Disassembly
Interleukin 12 Signaling
Reg. RNA Splicing
Reg. Actin Filament Length
Activation Of Nmda Receptors & Postsynaptic Events
Gnrh Signaling
Propanoate Metabolism
Establishment Of Protein Localization To Mito.
Insulin Receptor Signaling
Organelle Biogenesis & Maintenance
Positive Reg. Autophagy
Cellular Resp. to Insulin Stimulus
Reg. Nervous System Development
Valine Leucine & Isoleucine Degradation
Dephosphorylation
Negative Reg. Tor Signaling
Cognition
Resp. to Peptide Hormone
mTOR Signalling
Reg. Neurogenesis
Resp. to Insulin
Reg. Gen. of Precursor Metabolites
Establishment Of Protein Localization To Membrane
Long Term Potentiation
Regeneration
Lipid Homeostasis
−2
−1
0
1
signed −log10 p (positive: up)
GO_Slim
GOBP
KEGG_Legacy
Reactome
proteins
20
40
60
80
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_LR, fry sets at p < 0.05
73 sets. Page 2 of 2.
Protein DNA Complex Org.
Heme Metabolism
DNA Double Strand Break Repair
mRNA 3 End Processing
Pre−mRNA Processing
mRNA Polyadenylation
RNA Splicing Via Transesterification Reactions
Protein Depolymerization
mRNA Splicing
Negative Reg. Protein Containing Complex Disassembly
Negative Reg. mRNA Metabolism
Reg. Protein Depolymerization
Reg. mRNA Metabolism
Actin Filament Depolymerization
Negative Reg. RNA Catabolism
Telomere Org.
E2F Targets
Reg. mRNA Catabolism
Mitotic Prophase
Reg. DNA Binding Transcription Factor Activity
RNA Splicing
Reg. Telomere Maintenance Via Telomere Lengthening
Dengue Virus Host Interactions
Positive Reg. mRNA Metabolism
Reg. mRNA Processing
Negative Reg. mRNA Catabolism
Reg. mRNA Splicing Via Spliceosome
Actin Filament Polymerization
Negative Reg. Protein Polymerization
DNA Recombination
MYC Targets V1
0
1
2
3
signed −log10 p (positive: up)
proteins
30
60
90
120
GO_Slim
GOBP
Hallmark
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Baseline_HRvLR, fry sets at p < 0.05
18 sets. Page 1 of 1.
Reg. Resp. to Cytokine Stimulus
IL2 STAT5 Signaling
DNA Double Strand Break Repair
Signaling By Ntrks
Axo Dendritic Transport
Positive Reg. Cell Development
MT Based Transport
Gliogenesis
Reg. Cell Junction Assembly
Cytoskel. Dependent Intracell. Transport
Long Term Potentiation
Cell Substrate Junction Org.
Vascular Smooth Muscle Contraction
Schwann Cell Differentiation
Positive Reg. Phosph.
FA Metabolism
Protein Transmembrane Transport
Learning
−2
−1
0
1
2
3
signed −log10 p (positive: up)
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
15
20
25
30
35
40
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

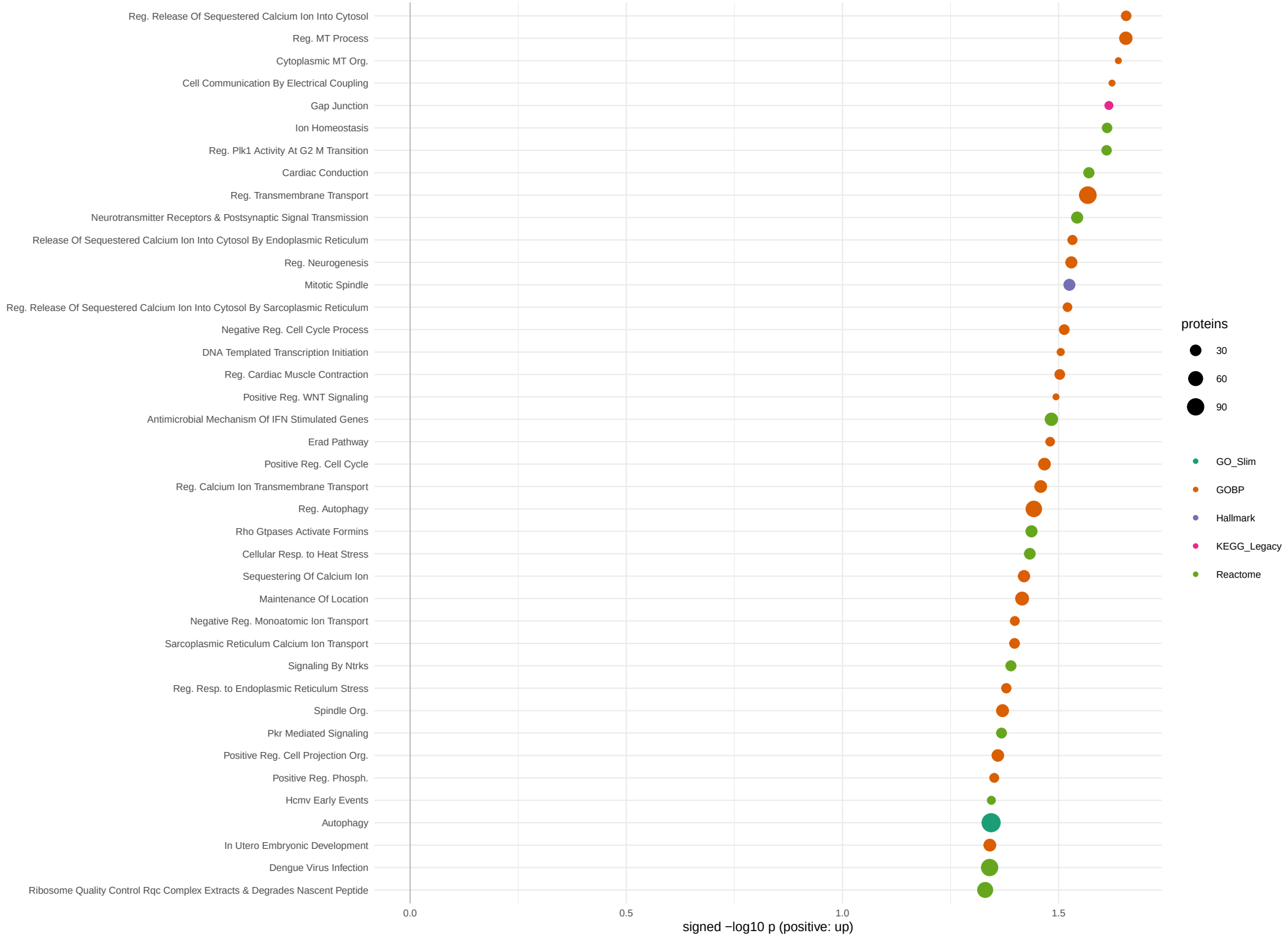

Trained_HRvLR, fry sets at p < 0.05
13 sets. Page 1 of 1.
Peptidyl AA Mod.
Cardiac Conduction
Ion Channel Transport
Ion Homeostasis
Resp. to Peptide Hormone
Positive Reg. Protein Secretion
Cellular Resp. to Insulin Stimulus
Resp. to Virus
Negative Reg. Growth
Insulin Receptor Signaling
Resp. to Insulin
Defense Resp. to Virus
Protein Transmembrane Transport
−2
−1
0
1
signed −log10 p (positive: up)
GOBP
Reactome
proteins
20
40
60
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.



fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

Acute_HRvLR, fry sets at p < 0.05

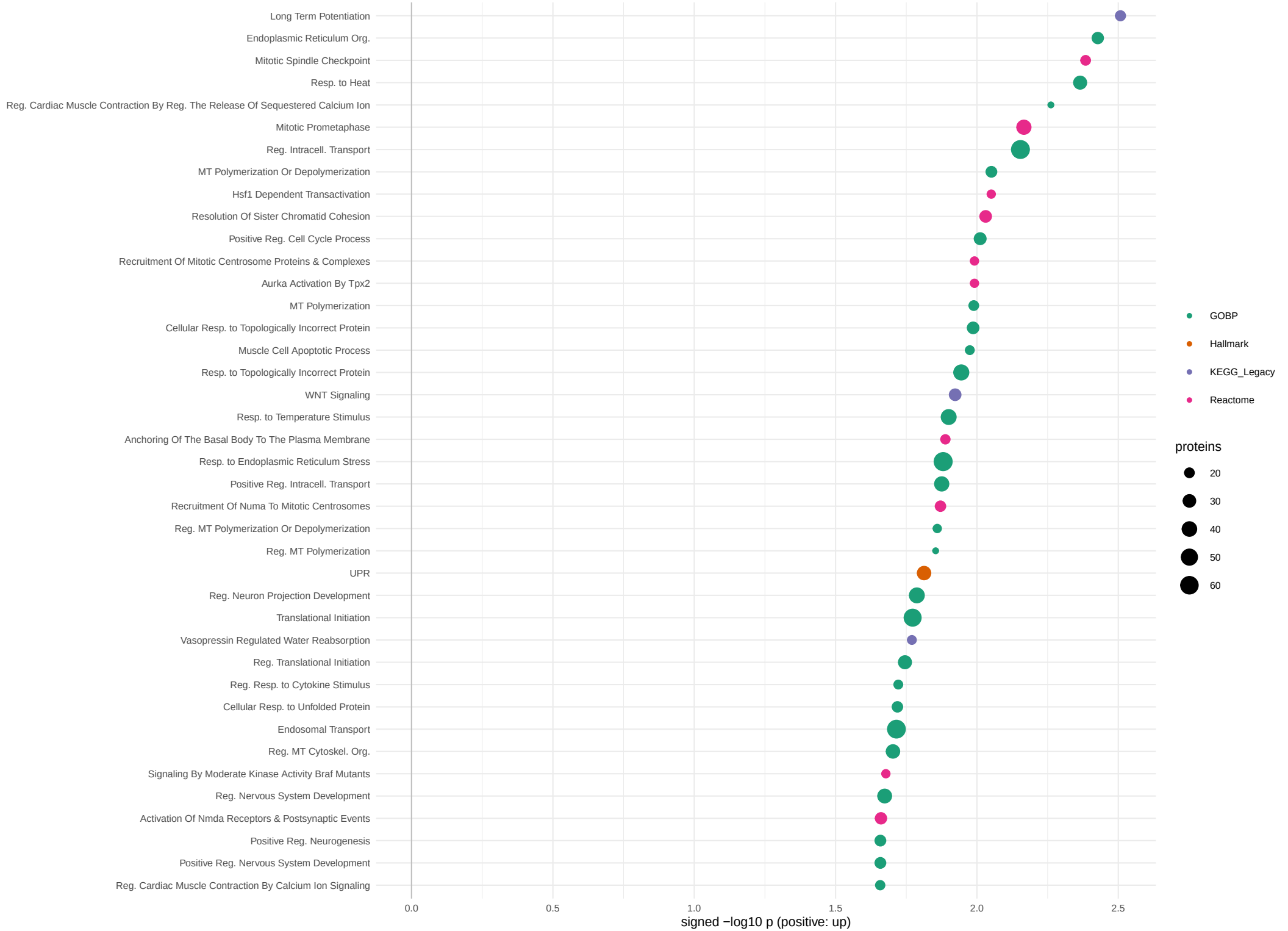
122 sets. Page 2 of 4.



fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

Acute_HRvLR, fry sets at p < 0.05

122 sets. Page 3 of 4.



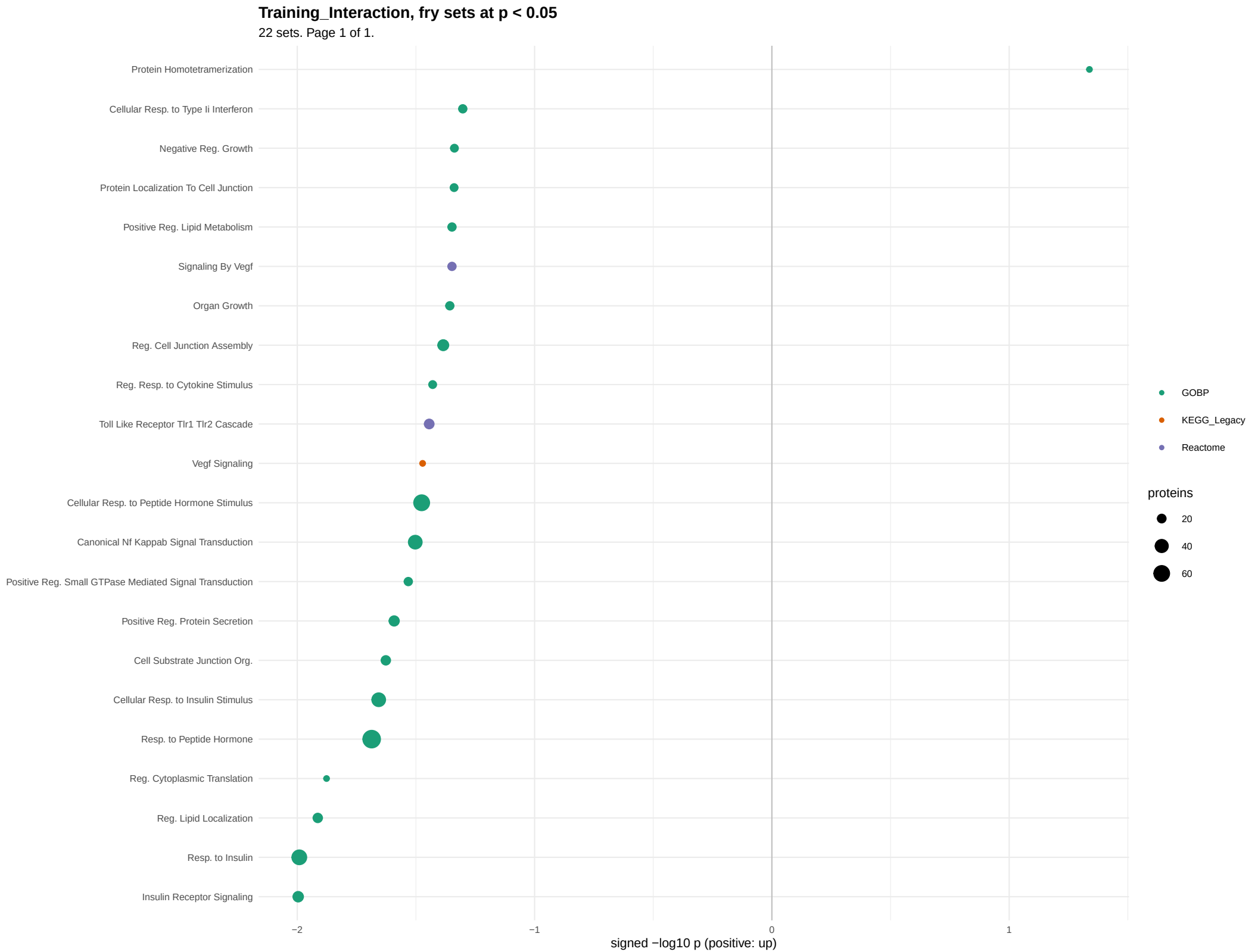
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

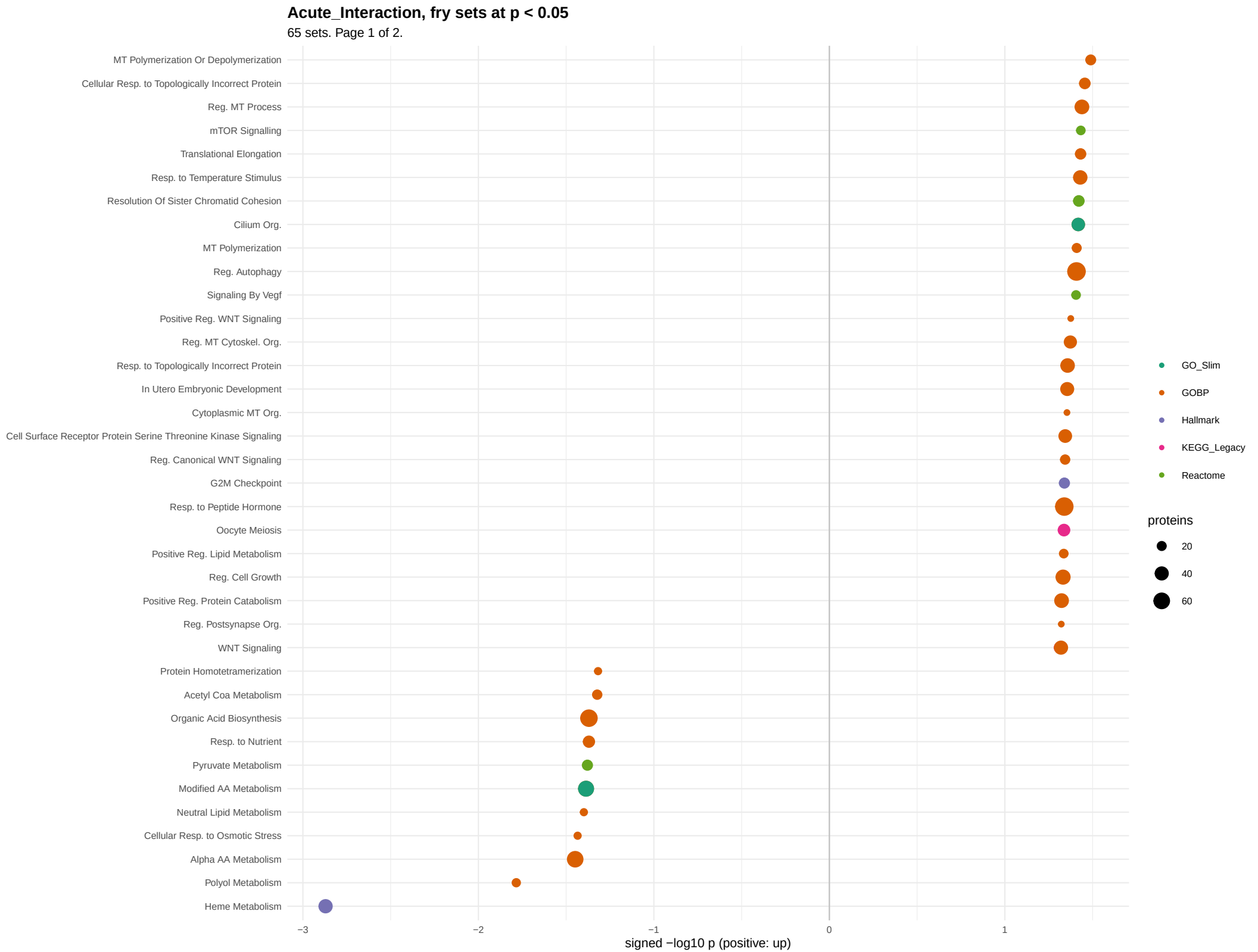
Acute_HRvLR, fry sets at p < 0.05

122 sets. Page 4 of 4.

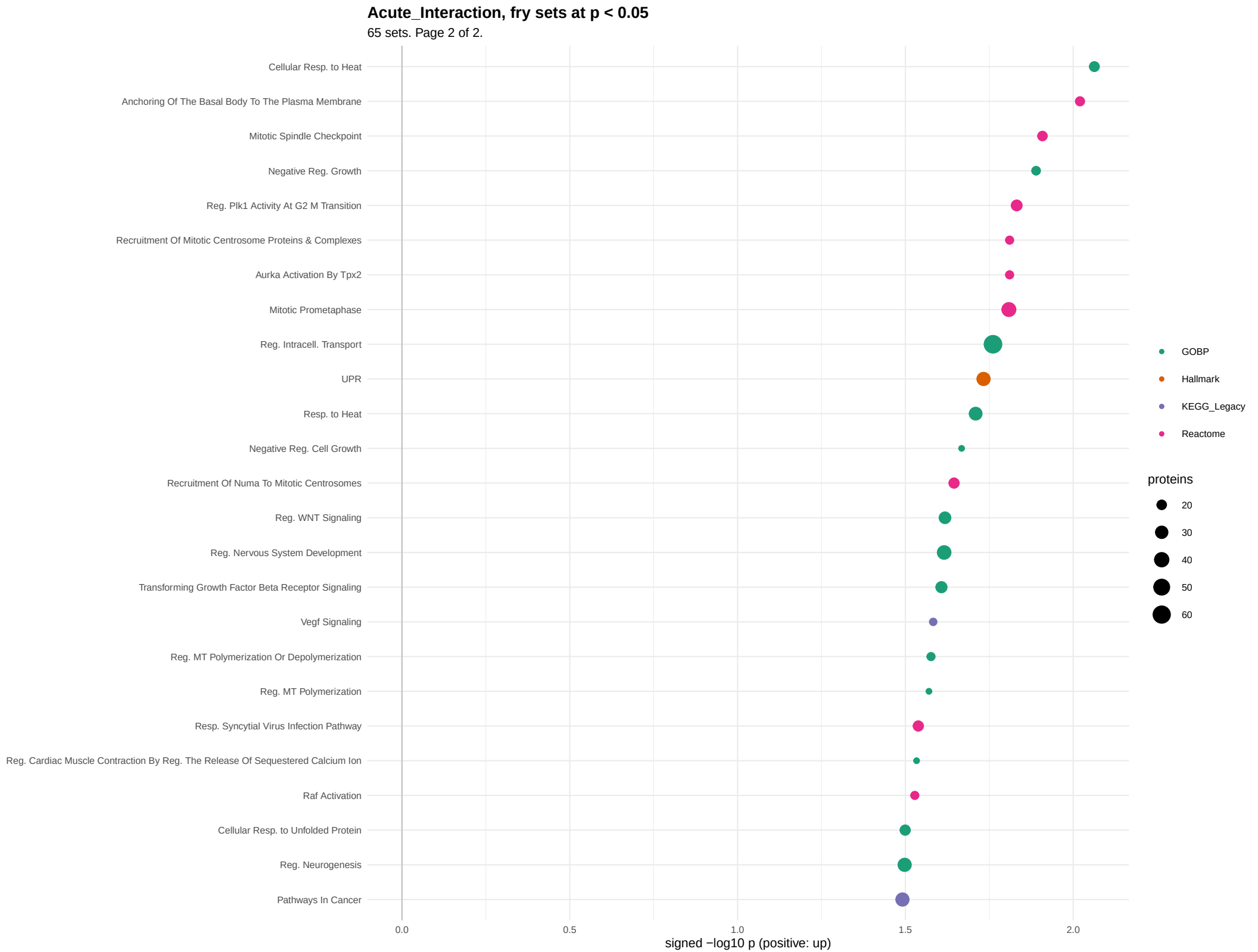


fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Training_Interaction, fry sets at p < 0.05
22 sets. Page 1 of 1.
Protein Homotetramerization
Cellular Resp. to Type Ii Interferon
Negative Reg. Growth
Protein Localization To Cell Junction
Positive Reg. Lipid Metabolism
Signaling By Vegf
Organ Growth
Reg. Cell Junction Assembly
Reg. Resp. to Cytokine Stimulus
Toll Like Receptor Tlr1 Tlr2 Cascade
Vegf Signaling
Cellular Resp. to Peptide Hormone Stimulus
Canonical Nf Kappab Signal Transduction
Positive Reg. Small GTPase Mediated Signal Transduction
Positive Reg. Protein Secretion
Cell Substrate Junction Org.
Cellular Resp. to Insulin Stimulus
Resp. to Peptide Hormone
Reg. Cytoplasmic Translation
Reg. Lipid Localization
Resp. to Insulin
Insulin Receptor Signaling
−2
−1
0
1
signed −log10 p (positive: up)
GOBP
KEGG_Legacy
Reactome
proteins
20
40
60
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.



fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_Interaction, fry sets at p < 0.05
65 sets. Page 2 of 2.
Cellular Resp. to Heat
Anchoring Of The Basal Body To The Plasma Membrane
Mitotic Spindle Checkpoint
Negative Reg. Growth
Reg. Plk1 Activity At G2 M Transition
Recruitment Of Mitotic Centrosome Proteins & Complexes
Aurka Activation By Tpx2
Mitotic Prometaphase
Reg. Intracell. Transport
UPR
Resp. to Heat
Negative Reg. Cell Growth
Recruitment Of Numa To Mitotic Centrosomes
Reg. WNT Signaling
Reg. Nervous System Development
Transforming Growth Factor Beta Receptor Signaling
Vegf Signaling
Reg. MT Polymerization Or Depolymerization
Reg. MT Polymerization
Resp. Syncytial Virus Infection Pathway
Reg. Cardiac Muscle Contraction By Reg. The Release Of Sequestered Calcium Ion
Raf Activation
Cellular Resp. to Unfolded Protein
Reg. Neurogenesis
Pathways In Cancer
0.0
0.5
1.0
1.5
2.0
signed −log10 p (positive: up)
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
20
30
40
50
60
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.